L07 — study notes

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Lecture 7 — Bayesian games, adverse selection and auctions · Map of the notes

Estimated unit. Built on 2026-10-16 from the study guide before the lecture (Tue Nov 17 2026). Reconcile with the slides when Aula/Lecture07.pdf arrives (/notes 07 –reconcile).

One claim. Private information is modelled as Nature drawing types from a commonly known distribution, and **a Bayesian Nash equilibrium is a Nash equilibrium of type-contingent plans**. **When one side's type enters the other's payoff, a single price selects the wrong half of the market**: in Tadelis's orange grove every unit is worth more to the buyer, yet only the worst quality trades. **Auctions are Bayesian games with a clean answer**: bid your value in a second-price auction, shade to $\frac{n-1}{n}\theta$ in a first-price one, and **expect the same revenue, $\frac{n-1}{n+1}$, either way**. **With common values, condition on the event in which your choice matters** — winning, being pivotal, being accepted — or be cursed.

Notation. Lecture 1's slides write $G = (I, \{S_i\}, \{u_i\})$; Tadelis Ch. 12 writes N and v_i. These notes write I, u_i, and Tadelis's types $\theta_i \in \Theta_i$ and beliefs ϕ_i.

Layer	What it means for this unit
U — intuition	Gibbons Ch. 3's prose and examples (Battle of the Sexes, linear auction, double auction); Cabral's insurer and auction stories; M&R's insurance and credit markets
C — course (examinable)	Tadelis Defs. 12.1–12.4, Claims 12.1–12.3, Prop. 12.1, Props. 13.1–13.2, Claim 13.1, eqs. (12.1)–(12.6), (13.1)–(13.10); Gibbons's Cournot and double-auction derivations
G — graduate	MWG Def. 8.E.1 / Prop. 8.E.1; MWG §13.B (competitive adverse selection, Example 13.B.1, constrained optimality); purification theorem; Myerson–Satterthwaite. **Named, not examinable** unless the lecture brings them

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Reading order

Part I — the machinery (Tadelis Ch. 12, Gibbons §3.1)

1. L07-01-bayesian-games-harsanyi — Ch. 12 opening, §12.1.1–12.1.2 (pp. 241–249): types, Nature, the **common prior**, Def. 12.1; the **entry game** turned into a 2×4 matrix with three equilibria at $p = 2/3$; posteriors by Bayes' rule (Def. 12.2) on the 1/6–1/3 prior; private vs common values.
2. L07-02-bayesian-nash-equilibrium — §12.1.3, §12.2.1 (pp. 249–255): a strategy maps types to actions (Def. 12.3); **BNE** (Def. 12.4, eq. 12.2) checked type by type; **chicken** solved in full, unique BNE (DC, dc) — and the opposite prediction when the payoffs are re-told.
3. L07-03-cournot-private-costs — Gibbons §3.1.A (pp. 144–146): the three FOCs solved step by step; **the high-cost type produces more, the low-cost type less** than under complete information; profits show who gains from the asymmetry.
4. L07-04-threshold-strategies-purification — §12.2.2, §12.5 (Tadelis pp. 255–258, 264–266); Gibbons §3.2.A (pp. 152–154): **best responses are cutoffs**; unique BNE $c^{1/3}$; **purification**: the Battle-of-the-Sexes cutoff BNE converges to the 2/3 mixed equilibrium.

Part II — information and trade (Tadelis §12.3, §12.4; Gibbons §3.2.C; applications)

5. L07-05-adverse-selection-lemons — §12.3 (pp. 258–261), M&R pp. 149–153, Cabral pp. 174–176, Gibbons pp. 158–163, MWG §13.B: **unravelling** to the lowest quality (Prop. 12.1); the premium-vs-spread condition $\Delta \geq (k-1)s/2$; M&R's **breakdown iff** $c\bar{x} > v$; Cabral's -2.55; the **double auction** derived (trade iff $v_b \geq v_s + \frac{1}{4}$, gains $\frac{9}{64}$ of $\frac{1}{6}$).

Part III — auctions (Tadelis Ch. 13, Gibbons §3.2.B)

6. L07-06-second-price-english — §13.1.1–13.1.2 (pp. 270–276): four formats, IPV; **Prop. 13.1** proved by the three cases; the button auction (Prop. 13.2); Cabral's auction vs posted price (125 vs 100). Builds on Lecture 1's dominance argument.
7. L07-07-first-price-dutch — §13.1.3 (pp. 276–279), Gibbons pp. 155–158: Claim 13.1; **(13.1)–(13.8) derived line by line**; the bid equals $E[Y \mid Y < \theta]$; Gibbons's $v/2$ two ways; **Dutch ≡ first-price**.
8. L07-08-revenue-equivalence — §13.1.4 (pp. 279–282): the four conditions; order statistics; $\frac{n-1}{n+1}$ by (13.9) and by (13.10); **equivalence type by type**; same mean, different risk.

Part IV — common values (Tadelis §12.4, §13.2)

9. L07-09-common-values-curses — §12.4 (pp. 261–264), §13.2 (pp. 282–284), Cabral p. 142: the **swing voter's curse** (pivotal posterior $= q$); the **winner's curse** (-1.25 for bidding 15); one table linking lemons, jury and auction.

Support

10. L07-10-formulas — the study-guide content mapped; every definition and number.
11. L07-11-pitfalls — 12 confusions with counterexamples; 26 original True/False items.
12. L07-12-python — every routine with asserts against the books' numbers.

The one result to hold

$$\boxed{s_i^*(\theta_i) \text{ maximizes } E_{\theta_{-i}}\left[u_i(a_i, s_{-i}^*(\theta_{-i}); \theta_i) \,\middle|\, \theta_i\right] \text{ for every type } \theta_i}$$

Every result of the lecture is this condition evaluated in one setting: a matrix (chicken), three FOCs (Cournot), a cutoff (study groups), a price the seller accepts (lemons), a bid (auctions). The economics lives in **what the expectation conditions on** — and with common values, the event "my action counts" is itself information.

Numbers and cases to have memorized

Case	Number	Meaning
Orange grove (p. 260)	pools worth **14, 19, 24** at prices 10, 20, 30	only L trades; BNE prices $[10, 14]$
Chicken (p. 254), $R = 8, H = 16, L = 0$	(DC, dc), payoffs $(3, 3)$	lenient drives, harsh swerves
Study groups	$\hat{\theta}^* = c^{1/3}$	free-riding cutoff
First-price, uniform	$s(\theta) = \frac{n-1}{n}\theta$; Gibbons $v/2$	shading θ/n falls with n
Revenue, uniform	$\frac{n-1}{n+1}$ ($n = 2$: $\frac{1}{3}$)	same in all four formats
Double auction	trade iff $v_b \geq v_s + \frac{1}{4}$; $\frac{9}{64}$ vs $\frac{1}{6}$	two-sided private information costs 15.6% of the gains
Oil field (p. 284)	$E[v \mid L] = 15$, $E[v \mid H] = 25$, payoff -1.25	the winner's curse
Jury	$\Pr\{G \mid \theta_I, \theta_G\} = q$	the swing voter's curse

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Figures

Built by `figures/make_figures.py` (run: `python Notas/L07/figures/make_figures.py`; it ends with asserts on a plotted number of every figure).

Figure	Shows	Lives in
L07-cournot-private-costs	best-response lines; one q_1^*, two q_2^*; complete-information points	L07-03-cournot-private-costs §5
L07-study-group-thresholds	(12.5) for both players, crossing at $c^{1/3}$	L07-04-threshold-strategies-purification §1.5
L07-lemons-unravelling	Tadelis's step vs the 45° line; MWG Example 13.B.1	L07-05-adverse-selection-lemons §4
L07-double-auction-regions	one-price vs linear trade regions	L07-05-adverse-selection-lemons §7.4
L07-spa-dominance	Fig. 13.1's regions as payoff curves	L07-06-second-price-english §2.1
L07-fpa-bidding	bid functions for $n = 2, 3, 5, 10$; payoff vs own bid	L07-07-first-price-dutch §4
L07-revenue-equivalence	the two price densities at $n = 3$; $\frac{n-1}{n+1}$ vs Monte Carlo	L07-08-revenue-equivalence §4

Interactive companions

- The Bid Shading Desk (`companions/companion-auction-bidding.html`) — move n, your value θ and your bid b; read the first-price bid $(n-1)\theta/n$ (the answer), the second-price bid θ, the expected payment and surplus of your type (equal in both formats), the revenue $(n-1)/(n+1)$, and what your deviation earns in each format. Linked from notes 06, 07, 08.
- The Lemons Ladder (`companions/companion-lemons.html`) — move the lowest value v_1, the quality spread s, the number of qualities K and the buyer premium Δ; read the highest quality that can trade, the BNE price band, the buyer's best offer and the gains lost. Defaults are Tadelis p. 258. Linked from note 05.

Both are checked by `companions/check_companions.py` (Node runs each page's model; Python recomputes it by another route, sweeps every control, and renders the page against a stub DOM).

Scope and ceiling

- **Examinable (C):** everything in Tadelis Ch. 12 and §13.1–13.2 used above; Gibbons's Cournot, linear-auction and double-auction derivations.
- **Named only (G):** MWG §8.E's formalism, §13.B's competitive equilibrium and constrained optimality; Harsanyi's purification theorem; the Mertens–Zamir universal type space; Myerson–Satterthwaite; English ≠ second-price with common values (Krishna, Milgrom).
- **Out (CLAUDE.md scope):** mechanism design — Tadelis Ch. 14, MWG Ch. 23, Gibbons §3.3 (the Revelation Principle). Revenue equivalence is **stated and checked for the uniform two-auction case**, not derived as a mechanism-design theorem; optimal reserve prices are left to Tadelis's exercises. Moral hazard and cheap talk are out (the grove's "my soil is good" remark is the only cheap talk here).
- Signaling and screening are the **next** lecture's subject.

Course-wide position

Lecture	Relation to this one
L01	weak dominance; **the second-price auction's truthful bid**, now a type-contingent strategy
L02	Nash equilibrium and mixed strategies; **a BNE is a Nash equilibrium** of the Harsanyi game; purification reinterprets the mix
L03	strategies as complete plans; the entry game's SPE; Nature as a player
L04	ultimatum bargaining — the orange-grove offer is a take-it-or-leave-it game with private information
L05, L06	reputation and relational contracts are the dynamic answer to hidden types; here types are static
L08	**signaling**: the informed side moves first and the uninformed update (PBE) — the way out of the lemons pool

Rules packs: Topic 7 (its adverse-selection section is a placeholder that note 05 can fill) · rules/game-theory (items 1, 5, 8, 9, 19).

Bibliography actually used

Book	Where
Tadelis (2013), Ch. 12–13 — the spine	every note
Gibbons (1992), Ch. 3	U layer; notes 03 (§3.1.A), 04 (§3.2.A), 05 (§3.2.C), 07 (§3.2.B)
MWG (1995) §8.E, §13.B	G layer of notes 01, 02, 04, 05
Milgrom & Roberts (1992), Ch. 5, pp. 149–153	note 05 §5 (insurance, credit rationing)
Cabral (2017), §6.4 and §7.4	notes 05 §6, 06 §4, 09 §3
Muñoz-Garcia & Toro-Gonzalez (2019), Ch. 7–8	drill, cited by exercise and page in every note

Related project material. Lecture 1's second-price auction: `Listas/lista-lecture-01.tex` Q1 b), `Listas/lista-lecture-01-check.py::spa_payoff`, `Listas/pratica-lecture-01.md` items 17, 23, 24 with `Listas/pratica-lecture-01-solutions/item17.tex`, `item23.tex`, `item24.tex`, `Simulados/quiz-lecture01-2-dominance-pareto-2026-10-13.md`. No list, quiz or video for Lecture 7 yet. **After class** (study guide): Muñoz-Garcia Ch. 7 (pp. 317–347), Ch. 8 (pp. 349–373); MWG 13.B.1, 13.B.2 (p. 473).

12 · 12.1.1–12.1.2 · Types, Nature and the common prior: incomplete information made playable

> **Claim.** A game of incomplete information cannot be solved as it stands: nobody knows the game. **Harsanyi's move** turns it into a game we already know how to solve: **Nature draws each player's type from a commonly known distribution, each player sees only his own type**, and the uncertainty about *payoffs* becomes uncertainty about *Nature's move* — a game of imperfect information. Everything after this note is Nash equilibrium applied to that game.

Prev: — · Next: L07-02-bayesian-nash-equilibrium · Map: L07-00-map

Notation. The Lecture 1 slides write a game as $G = (I, \{S_i\}_{i\in I}, \{u_i\}_{i\in I})$; Tadelis Ch. 12 writes the player set N and payoffs v_i. These notes keep the slides' I and u_i and Tadelis's $\theta_i \in \Theta_i$ (types) and ϕ_i (beliefs). Gibbons writes $t_i \in T_i$ and p_i.

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1 · U — what complete information assumed

Every game so far assumed that **the game itself is common knowledge**: who plays, what each can do, and how outcomes map into payoffs (Lecture 1, Def. 3.1 p. 45). Tadelis opens Ch. 12 (p. 241) with the Cournot duopoly: is it reasonable that each firm knows the other's cost function exactly? Gibbons opens Ch. 3 (p. 143) with the **sealed-bid auction**: each bidder knows her own valuation and not the others'.

> **Incomplete information = at least one player is uncertain about another player's payoff function** (Gibbons p. 143). It is a statement about *preferences*, not about moves.

The problem: to best-respond, player i needs a belief about what j will do; what j does depends on j's payoffs; i does not know them. And j needs a belief about i's belief about j's payoffs, and so on. Harsanyi cut the infinite regress with one modelling device.

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2 · C — Harsanyi's transformation, worked on Tadelis's entry game

2.1 · The complete-information game (Fig. 12.1, p. 243)

Player 1 (entrant) chooses Out (O) or Enter (E); after E, player 2 (incumbent) chooses Fight (F) or Accommodate (A).

Outcome	u_1	u_2
O	0	2
E, then F	-1	-1
E, then A	1	1

Backward induction: after E, player 2 compares -1 with 1 and accommodates, so player 1 enters. The unique SPE is (E, A); (O, F) is a Nash equilibrium sustained by an incredible threat.

2.2 · Add a second type of incumbent (Fig. 12.2, p. 243)

Now player 2 is **rational** (payoffs as above) with probability p, or **crazy** with probability $1-p$: the crazy type *enjoys* fighting, and (E, F) pays him 2 instead of -1. Everything else is unchanged.

The three things that must be pinned down (pp. 243–244):

1. **Nature moves first** and picks player 2's type.
2. **Player 2 knows his own type** when he moves; player 1 does not — hence a single information set for player 1 across Nature's two branches.
3. **The probability p is common knowledge** — the **common prior assumption**: everyone agrees on how Nature draws types. *"This is of course a strong assumption, but without it we will not be able even to discuss equilibrium behavior"* (p. 244).

2.3 · The induced normal form

A strategy for player 2 must say what **each type** does: $xy \in \{AA, AF, FA, FF\}$, with x the rational type's action and y the crazy type's. Player 1 has $\{O, E\}$. Each strategy pair produces a lottery over outcomes, and expected payoffs fill the matrix. For (E, AF):

$$u_1 = p \cdot 1 + (1-p) \cdot (-1) = 2p - 1, \qquad u_2 = p \cdot 1 + (1-p) \cdot 2 = 2 - p.$$

For (E, FA) each type plays the action he likes less (rational fights, crazy accommodates): $u_1 = p(-1) + (1-p)(1) = 1 - 2p$ and $u_2 = p(-1) + (1-p)(1) = 1 - 2p$. For (E, FF): $u_1 = -1$, $u_2 = p(-1) + (1-p)2 = 2 - 3p$. After O every cell is $(0, 2)$.

	AA	AF	FA	FF
O	$0, 2$	$0, 2$	$0, 2$	$0, 2$
E	$1, 1$	$2p-1,\ 2-p$	$1-2p,\ 1-2p$	$-1,\ 2-3p$

At $p = 2/3$ (the book's choice, p. 244):

	AA	AF	FA	FF
O	$0, 2$	$0, 2$	$0, 2$	$0, 2$
E	$1, 1$	$\frac{1}{3}, \frac{4}{3}$	$-\frac{1}{3}, -\frac{1}{3}$	$-1, 0$

Solve it like any bimatrix game. Player 1's best responses: against AA or AF enter ($1 > 0$, $\frac{1}{3} > 0$); against FA or FF stay out ($0 > -\frac{1}{3}$, $0 > -1$). Player 2's best responses: against O every column pays 2 (all are best responses); against E the column payoffs are $1, \frac{4}{3}, -\frac{1}{3}, 0$, so only AF. Mutual best responses:

$$\boxed{(O, FA), \quad (O, FF), \quad (E, AF)} \qquad \text{(p. 245)}$$

Only (E, AF) survives the extensive-form check that each type moves optimally at his own node (rational accommodates, crazy fights); the other two rest on types who would not carry out F or A.

2.4 · Why the "meta-player" gets it right (p. 245)

The matrix averages player 2's types into one player who maximizes the **prior-weighted** sum of his types' payoffs. That meta-player's best response coincides with each type's best response **because a strategy specifies an action per type**: against E, moving from AA to AF changes only the crazy type's action and raises only his payoff, so the average rises. AA is not a best response to E precisely because it asks the crazy type to play a dominated action.

> **The device is not free.** Before, equilibrium asked players to hold correct beliefs about *actions*. Now it asks, in addition, that **everyone agree on how types are drawn** (p. 245–246). Predictions are only as good as that leap.

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3 · C — the general object: Definition 12.1 (p. 246)

> **Definition 12.1.** The normal-form representation of an n-player static Bayesian game of incomplete information is
>
> $$\langle I, \{A_i\}_{i=1}^n, \{\Theta_i\}_{i=1}^n, \{u_i(\cdot;\theta_i), \theta_i \in \Theta_i\}_{i=1}^n, \{\phi_i\}_{i=1}^n \rangle$$
>
> where A_i is i's action set; $\Theta_i = \{\theta_{i1}, \dots, \theta_{ik_i}\}$ is i's **type space**; $u_i : A \times \Theta_i \to \mathbb{R}$ is the **type-dependent payoff**; and $\phi_i(\theta_{-i} \mid \theta_i)$ is i's (posterior) belief about the other players' types given his own.

Timing (pp. 246–247), the four steps every Bayesian game in this lecture follows:

Step	What happens
1	Nature draws a type profile $(\theta_1, \dots, \theta_n)$
2	Each i learns **only** θ_i and forms $\phi_i(\theta_{-i} \mid \theta_i)$ from the prior
3	Players choose $a_i \in A_i$ **simultaneously** (static game)
4	Payoffs $u_i(a;\theta_i)$ are realized

3.1 · Private values versus common values

	Payoff depends on	Examples in this lecture
Private values	own type only: $u_i(a;\theta_i)$	chicken, study groups, IPV auctions (L07-06-second-price-english)
Common values	others' types too: $u_i(a;\theta_1, \dots, \theta_n)$	the orange grove (L07-05-adverse-selection-lemons), the jury and the oil field (L07-09-common-values-curses)

> **Every "curse" in this lecture — adverse selection, the swing voter's curse, the winner's curse — needs common values.** With private values, learning another player's type changes what you expect him to *do*, never what the outcome is *worth to you*.

Gibbons makes the same extension in two lines (p. 149): when one player's private information enters another's payoff, write $u_i(a_1, \dots, a_n; t_1, \dots, t_n)$.

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4 · C — beliefs from the common prior: Definition 12.2 (pp. 247–249)

Each player learns his own type; if types are **correlated**, his type is evidence about the others'. Bayes' rule converts the common prior into his interim belief.

> **Definition 12.2.** Conditional on event S, the probability of event H is
>
> $$\Pr\{H \mid S\} = \frac{\pi(S \cap H)}{\pi(S)}. \tag{12.1}$$

4.1 · Worked — the book's correlated prior (p. 249)

Types $\theta_1 \in \{a, b\}$, $\theta_2 \in \{c, d\}$, prior:

	$\theta_2 = c$	$\theta_2 = d$	row sum
$\theta_1 = a$	1/6	1/3	1/2
$\theta_1 = b$	1/3	1/6	1/2

Player 1 learns $\theta_1 = a$:

$$\phi_1(c \mid a) = \frac{1/6}{1/6 + 1/3} = \frac{1/6}{1/2} = \frac{1}{3}, \qquad \phi_1(d \mid a) = \frac{1/3}{1/2} = \frac{2}{3}.$$

Had he learned b: $\phi_1(c \mid b) = \frac{1/3}{1/2} = \frac{2}{3}$, $\phi_1(d \mid b) = \frac{1}{3}$. **His type moved his belief**: before learning it, $\Pr\{c\} = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$.

Independent types (every application in Gibbons §3.2, p. 148): the prior factors, $\pi(\theta_1, \theta_2) = \pi_1(\theta_1)\pi_2(\theta_2)$, so $\phi_i(\theta_{-i} \mid \theta_i) = \pi_{-i}(\theta_{-i})$ — the posterior equals the prior and does not depend on one's own type. Then the other players also *know* i's belief (Gibbons p. 149).

> **Correlation is where conditioning bites.** In the oil-field auction of L07-09-common-values-curses the signals are correlated, and it is exactly the update "I won, so the other signal was low" that bidders who ignore Bayes' rule get wrong.

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5 · G — formal side (not examinable unless the lecture brings it)

- **The belief hierarchy.** A type must encode beliefs about beliefs about … payoffs. Harsanyi's argument that a rich enough type space captures the whole hierarchy was made precise by Mertens and Zamir (1985) and Brandenburger and Dekel (1993) (Tadelis's Remark, p. 247). Named only.
- **MWG §8.E** (p. 255) writes a Bayesian game as $[I, \{S_i\}, \{u_i(\cdot)\}, \Theta, F(\cdot)]$, with the joint distribution F of the types as the common prior; Example 8.E.1 (p. 254) is a two-type prisoner's dilemma solved exactly like §2 above.
- **Gibbons's version** (p. 148): $G = \{A_1, \dots, A_n; T_1, \dots, T_n; p_1, \dots, p_n; u_1, \dots, u_n\}$, beliefs by Bayes' rule $p_i(t_{-i} \mid t_i) = p(t_{-i}, t_i) / \sum_{t'_{-i}} p(t'_{-i}, t_i)$ (p. 149). Different letters, same object.

Links

- Next: strategies as type-contingent plans and the equilibrium concept → L07-02-bayesian-nash-equilibrium.
- The entry game's complete-information version and SPE: Lecture 3.
- Common knowledge (Def. 3.1) and the normal form (Def. 3.3): Lecture 1.
- Rules: rules/game-theory items 1, 5, 9.
- Code: L07-12-python §1 (`harsanyi_normal_form`, `posterior`).
- **Exercises (number + page only):** Tadelis 12.1 (p. 266); Gibbons 3.1 (p. 169); Muñoz-Garcia Ch. 7 Ex. 1 (p. 318), Ex. 2 (p. 323).

12.1.3 · 12.2.1 · A strategy is a plan for every type; a BNE is a Nash equilibrium of those plans

> **Claim.** In a Bayesian game a **strategy maps types to actions**, and a **Bayesian Nash equilibrium** is a profile of such maps in which **every type of every player** best-responds to his posterior over the others' types and their strategies. Checking it type by type is all the work; the chicken game shows the method end to end and gives a prediction ("children of lenient parents drive on") that **flips when the payoffs are reassigned**.

Prev: L07-01-bayesian-games-harsanyi · Next: L07-03-cournot-private-costs · Map: L07-00-map

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1 · C — strategies: Definition 12.3 (p. 249)

> **Definition 12.3.** A **pure strategy** for player i is a function $s_i : \Theta_i \to A_i$ that specifies the action $s_i(\theta_i)$ he chooses when his type is θ_i. A mixed strategy is a probability distribution over pure strategies.

Counting. With $|A_i|$ actions and $|\Theta_i|$ types, player i has $|A_i|^{|\Theta_i|}$ pure strategies: the entry game's incumbent, 2 actions $\times$ 2 types $\to 2^2 = 4$ (AA, AF, FA, FF); the chicken players below, likewise 4 each.

Why a plan for types that were not drawn (Tadelis pp. 250–251; Gibbons pp. 150–151): once Nature has drawn θ_i, the other types of i never move. But j **cannot compute his expected payoff without knowing what every type of i would do**, weighted by ϕ_j. A strategy that left some types unspecified would make j's best response undefined — the same reason a strategy in an extensive game must cover unreached information sets (Lecture 3).

> **Types are information sets.** Learning one's type is arriving at an information set that Nature's move determines (p. 250). A type-contingent strategy is a complete contingent plan, exactly as in Lecture 1's definition of a strategy.

A pure strategy of i looks mixed to j (p. 250): if the incumbent plays AF and is rational with probability p, the entrant faces "A with probability p, F with probability $1-p$". This is the seed of Harsanyi's reading of mixed strategies (L07-04-threshold-strategies-purification §3).

Gibbons's vocabulary (p. 150), needed in Lecture 8: a **separating** strategy gives each type a different action; a **pooling** strategy gives all types the same action.

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2 · C — Bayesian Nash equilibrium: Definition 12.4 (p. 251)

> **Definition 12.4.** A strategy profile $s = (s_1(\cdot), \ldots, s_n(\cdot))$ is a **pure-strategy Bayesian Nash equilibrium** if, for every player i, for each of i's types $\theta_i \in \Theta_i$, and for every $a_i \in A_i$,

$$\sum_{\theta_{-i} \in \Theta_{-i}} \phi_i(\theta_{-i} \mid \theta_i)\, u_i(s_i(\theta_i), s_{-i}(\theta_{-i}); \theta_i) \;\geq\; \sum_{\theta_{-i} \in \Theta_{-i}} \phi_i(\theta_{-i} \mid \theta_i)\, u_i(a_i, s_{-i}(\theta_{-i}); \theta_i). \quad (12.2)$$

Read (12.2) left to right:

Piece	Meaning
$\phi_i(\theta_{-i} \mid \theta_i)$	what type θ_i believes about the others' types (from the common prior)
$s_{-i}(\theta_{-i})$	what each of those types does — the others' **strategies**
left side	type θ_i's expected payoff from his prescribed action
right side	his expected payoff from any single alternative action

Two simplifications the book makes explicit (p. 251):

1. It suffices to check deviations to **actions**, one type at a time, not to whole alternative strategies $s_i'(\cdot)$: if no type gains by switching his action, no rearrangement across types gains either.
2. The general form, valid for a continuum of types (Remark, p. 251):

$$E_{\theta_{-i}}[u_i(s_i(\theta_i), s_{-i}(\theta_{-i}); \theta_i) \mid \theta_i] \;\geq\; E_{\theta_{-i}}[u_i(a_i, s_{-i}(\theta_{-i}); \theta_i) \mid \theta_i] \quad \forall a_i \in A_i.$$

A BNE is a Nash equilibrium of the Harsanyi game (Gibbons p. 151: *"a Bayesian Nash equilibrium is simply a Nash equilibrium in a Bayesian game"*). Two equivalent ways to find one: **ex ante** — build the matrix of prior-expected payoffs over type-contingent strategies and find its Nash equilibria (the entry game, L07-01-bayesian-games-harsanyi §2.3); **interim** — check (12.2) for each type separately. MWG Prop. 8.E.1 (p. 256) is the statement that the two coincide for every type that occurs with positive probability.

Existence (Gibbons pp. 151–152): a finite static Bayesian game has a BNE, possibly in mixed strategies — the proof is Nash's existence theorem applied to the Harsanyi game.

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3 · Worked — chicken with incomplete information (§12.2.1, pp. 252–255)

3.1 · The game

Two teenagers drive at each other; each chooses chicken (C/c) or drive (D/d).

Outcome	Payoff to i
i chickens	0 (whatever j does)
i drives, j chickens	R (all the respect)
both drive	$R/2 - k_i$ (split respect, accident cost k_i)

k_i is i's **type**: harsh parents H or lenient parents $L < H$, each with probability $\frac{1}{2}$, **independently** across players. Each kid knows his own parents only. Four equally likely states $\theta_1\theta_2 \in \{LL, LH, HL, HH\}$ (Fig. 12.3, p. 253). Strategies: $xy \in \{CC, CD, DC, DD\}$ for player 1, x = action of type L, y = action of type H; likewise $\{cc, cd, dc, dd\}$ for player 2.

3.2 · One cell by hand: (CD, dd) (p. 254)

Player 1's L-type chickens, H-type drives; both types of player 2 drive. Average over the four states, each with weight $\frac{1}{4}$:

$$Eu_1 = \tfrac{1}{4}\underbrace{u_1(C,d;L)}_{0} + \tfrac{1}{4}\underbrace{u_1(C,d;L)}_{0} + \tfrac{1}{4}\underbrace{u_1(D,d;H)}_{R/2-H} + \tfrac{1}{4}\underbrace{u_1(D,d;H)}_{R/2-H} = \frac{R}{4} - \frac{H}{2}.$$

$$Eu_2 = \tfrac{1}{4}\underbrace{u_2(C,d;L)}_{R} + \tfrac{1}{4}\underbrace{u_2(C,d;H)}_{R} + \tfrac{1}{4}\underbrace{u_2(D,d;L)}_{R/2-L} + \tfrac{1}{4}\underbrace{u_2(D,d;H)}_{R/2-H} = \frac{3R}{4} - \frac{L}{4} - \frac{H}{4}.$$

3.3 · The whole matrix at $R = 8$, $H = 16$, $L = 0$ (p. 254)

P1 \ P2	cc	cd	dc	dd
CC	$0, 0$	$0, 4$	$0, 4$	$0, 8$
CD	$4, 0$	$-1, -1$	$-1, 3$	$-6, 2$
DC	$4, 0$	$3, -1$	$\mathbf{3, 3}$	$2, 2$
DD	$8, 0$	$2, -6$	$2, 2$	$-4, -4$

Check against §3.2: (CD, dd) gives $8/4 - 16/2 = -6$ and $3 \cdot 8/4 - 0 - 16/4 = 2$. ✓

Best responses. Player 1's best row in each column: $cc \to DD$ (8); $cd \to DC$ (3); $dc \to DC$ (3); $dd \to DC$ (2). Player 2's best column in each row: $CC \to dd$ (8); $CD \to dc$ (3); $DC \to dc$ (3); $DD \to dc$ (2). The only cell that is a best response both ways:

$$\boxed{(DC,\ dc)} \quad \text{— the unique pure-strategy BNE (p. 254).}$$

3.4 · The same answer type by type (the interim check)

Take player 2 playing dc (lenient drives, harsh chickens). Player 1 believes each with probability $\frac{1}{2}$.

Player 1's type	C	D	Best
L $(k = 0)$	0	$\frac{1}{2}\underbrace{(4-0)}_{\text{vs. } d} + \frac{1}{2}\underbrace{8}_{\text{vs. } c} = 6$	D
H $(k = 16)$	0	$\frac{1}{2}(4-16) + \frac{1}{2}\cdot 8 = -2$	C

So DC is player 1's best reply to dc, and by symmetry dc to DC. The ex-ante payoff is $\frac{1}{2}\cdot 6 + \frac{1}{2}\cdot 0 = 3$, the matrix's $(3, 3)$. ✓

> **The prediction is only as good as the payoffs** (p. 255). With $R = 8$, $H = 16$, $L = 0$, "harsh upbringing" yields fewer crashes. Tell the story the other way — lenient parents' children respect the car ($L = 16$), harsh parents' children do not ($H = 0$) — and the unique BNE flips to (CD, cd). Same game form, opposite policy advice.

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4 · U — reading a BNE

- **Each type is effectively a separate player** maximizing against his conditional distribution of the others' actions (MWG's gloss on Prop. 8.E.1, p. 256).
- **Uncertainty about the opponent's type acts like a mixed strategy by the opponent.** Player 1 of type L above faces "drive with probability $\frac{1}{2}$" although player 2 plays a pure strategy.

- **A BNE is a consistency condition**, not a learning story: beliefs about types are given by the prior, beliefs about strategies must be correct (rules/game-theory item 17).

Links

- Prev: the Harsanyi transformation and the entry game → L07-01-bayesian-games-harsanyi.
- Next: the same definition with a continuum of actions → L07-03-cournot-private-costs.
- Nash equilibrium of a complete-information game: Lecture 2.
- Separating vs pooling strategies, and what changes when actions are observed before others move: Lecture 8.
- Code: L07-12-python §2 (`pure_nash` on the chicken matrix, both parameter sets).
- **Exercises (number + page only):** Tadelis 12.1 (p. 266), 12.3 (pp. 266–267), 12.4 (p. 267); Gibbons 3.4 (pp. 169–170); MWG 8.E.1 (p. 265); Muñoz-Garcia Ch. 7 Ex. 3 (p. 327), Ex. 4 (p. 331), Ex. 7 (p. 343).

Gibbons 3.1.A · Cournot with a privately known cost: one quantity for firm 1, two for firm 2

Claim. When firm 2 alone knows its marginal cost, its strategy is **a pair of quantities** $(q_2^*(c_H), q_2^*(c_L))$ and firm 1 best-responds to their **expectation**. Solving three first-order conditions gives the BNE in closed form, and it differs from complete-information Cournot in a definite direction: **the high-cost type produces more, the low-cost type less**, because each type also responds to firm 1 *not* knowing which one it is.

Prev: L07-02-bayesian-nash-equilibrium · Next: L07-04-threshold-strategies-purification · Map: L07-00-map

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1 · C — the game as a static Bayesian game (Gibbons pp. 144, 147)

Element	Here
Players	firms 1 and 2
Actions	$q_1 \geq 0$, $q_2 \geq 0$, chosen simultaneously
Types	$T_1 = \{c\}$ (firm 1's cost is common knowledge); $T_2 = \{c_H, c_L\}$, $c_L < c_H$
Prior	$\Pr\{c_H\} = \theta$, $\Pr\{c_L\} = 1 - \theta$, common knowledge
Payoffs	$\pi_1 = [(a - q_1 - q_2) - c]q_1$; $\pi_2(q_1, q_2; c_k) = [(a - q_1 - q_2) - c_k]q_2$
Strategies	firm 1: a number q_1; firm 2: a **function** $c_k \mapsto q_2(c_k)$, i.e. a pair

Demand $P(Q) = a - Q$, $Q = q_1 + q_2$. Firm 2 could be a new entrant or have a new technology (p. 144). *Gibbons writes θ for the probability of the high cost — not a type.*

Why firm 2's strategy must be a pair (Gibbons pp. 150–151): firm 1 cannot check that q_1 is a best response unless it knows what firm 2 would produce **at each cost**. A single number for firm 2 would leave firm 1's expected profit undefined.

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2 · C — the three optimization problems and their FOCs (p. 145)

Firm 2, high cost:

$$\max_{q_2} \left[(a - q_1^* - q_2) - c_H\right] q_2 \quad \Rightarrow \quad a - q_1^* - 2q_2 - c_H = 0 \quad \Rightarrow \quad q_2^*(c_H) = \frac{a - q_1^* - c_H}{2}.$$

Firm 2, low cost:

$$q_2^*(c_L) = \frac{a - q_1^* - c_L}{2}.$$

Firm 1 does not know which type it faces, so it maximizes **expected** profit:

$$\max_{q_1} \ \theta[(a - q_1 - q_2^*(c_H)) - c]q_1 + (1 - \theta)[(a - q_1 - q_2^*(c_L)) - c]q_1.$$

Differentiate:

$$\theta\,[a - 2q_1 - q_2^*(c_H) - c] + (1-\theta)[a - 2q_1 - q_2^*(c_L) - c] = 0$$

$$\Rightarrow \quad q_1^* = \frac{\theta[a - q_2^*(c_H) - c] + (1-\theta)[a - q_2^*(c_L) - c]}{2} = \frac{a - c - E[q_2^*]}{2},$$

with $E[q_2^*] = \theta q_2^*(c_H) + (1-\theta) q_2^*(c_L)$.

> **Firm 1's problem is linear in q_2, so only the mean of firm 2's quantity matters to it.** That is special to linear demand (see §5).

---

3 · C — solving, every step

Write $\bar{c} = \theta c_H + (1-\theta) c_L$ for firm 2's expected cost. Average firm 2's two FOCs with weights $\theta, 1-\theta$:

$$E[q_2^*] = \frac{a - q_1^* - \bar{c}}{2}.$$

Substitute into firm 1's FOC:

$$q_1^* = \frac{a - c - \frac{1}{2}(a - q_1^* - \bar{c})}{2} \;\Rightarrow\; 2q_1^* = \tfrac{a}{2} - c + \tfrac{q_1^*}{2} + \tfrac{\bar{c}}{2} \;\Rightarrow\; \tfrac{3}{2} q_1^* = \tfrac{a - 2c + \bar{c}}{2}$$

$$\boxed{q_1^* = \frac{a - 2c + \theta c_H + (1-\theta) c_L}{3}}$$

Back into firm 2's high-cost FOC, using $\bar{c} = c_H - (1-\theta)(c_H - c_L)$:

$$q_2^*(c_H) = \frac{a - c_H}{2} - \frac{a - 2c + \bar{c}}{6} = \frac{2a + 2c - \bar{c} - 3c_H}{6} = \frac{2a + 2c - 4c_H + (1-\theta)(c_H - c_L)}{6}$$

$$\boxed{q_2^*(c_H) = \frac{a - 2c_H + c}{3} + \frac{1-\theta}{6}(c_H - c_L)}$$

Likewise, with $\bar{c} = c_L + \theta(c_H - c_L)$:

$$\boxed{q_2^*(c_L) = \frac{a - 2c_L + c}{3} - \frac{\theta}{6}(c_H - c_L)}$$

(Gibbons assumes the FOCs characterize the solutions, i.e. all three quantities positive; he leaves a sufficient condition as an exercise on p. 145.)

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4 · C — comparison with complete information (p. 146)

Under complete information with costs c_1, c_2, firm i produces $q_i = (a - 2c_i + c_j)/3$.

Firm 2's cost	Complete information q_2	BNE q_2^*	Direction
c_H	$(a - 2c_H + c)/3$	$+\frac{1-\theta}{6}(c_H - c_L)$ more	**produces more**
c_L	$(a - 2c_L + c)/3$	$-\frac{\theta}{6}(c_H - c_L)$ less	**produces less**

Firm 2's cost	Complete information q_2	BNE q_2^*	Direction

Why. Firm 2 "not only tailors its quantity to its cost but also responds to the fact that firm 1 cannot do so" (p. 146). Facing a high-cost rival, firm 1 *would* expand; not knowing, it produces the average, which is **less** than it would against a known high-cost firm — so the high-cost type, facing less rival output than under complete information, produces more. The low-cost type faces more and produces less.

The "potentially misleading feature" (p. 146): here q_1^* equals the average of firm 1's two complete-information quantities,

$$\theta\frac{a-2c+c_H}{3}+(1-\theta)\frac{a-2c+c_L}{3}=\frac{a-2c+\bar{c}}{3}=q_1^*,$$

but only because costs enter linearly. Gibbons's counterexample: total cost $c_i q_i^2$.

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5 · Worked — an illustrative calibration

Gibbons states the model in letters only. Take $a=10$, $c=2$, $c_H=3$, $c_L=1$, $\theta=\frac{1}{2}$ (illustrative), so $\bar{c}=2$:

	Formula	Value
q_1^*	$(10-4+2)/3$	$8/3\approx 2.667$
$q_2^*(c_H)$	$(10-6+2)/3+\frac{1/2}{6}\cdot 2=2+\frac{1}{6}$	$13/6\approx 2.167$
$q_2^*(c_L)$	$(10-2+2)/3-\frac{1/2}{6}\cdot 2=\frac{10}{3}-\frac{1}{6}$	$19/6\approx 3.167$
$E[q_2^*]$	$\frac{1}{2}(13/6+19/6)$	$8/3$

Check each FOC. $(10-8/3-3)/2=(13/3)/2=13/6$ ✓; $(10-8/3-1)/2=(19/3)/2=19/6$ ✓; $(10-2-8/3)/2=(16/3)/2=8/3$ ✓.

Complete-information benchmarks. Against c_H: $q_1=(10-4+3)/3=3$, $q_2=(10-6+2)/3=2$. Against c_L: $q_1=(10-4+1)/3=7/3$, $q_2=(10-2+2)/3=10/3$. Indeed $13/6>2$ and $19/6<10/3$, and $q_1^*=\frac{1}{2}(3+7/3)=8/3$.

Profits. With linear demand each FOC says *margin = quantity* (e.g. $a-q_1-q_2^*(c_H)-c_H=q_2^*(c_H)$), so each profit is a square:

	BNE	Complete information	Who gains from the asymmetry
Firm 1 (expected)	$(8/3)^2=64/9\approx 7.111$	$\frac{1}{2}\cdot 3^2+\frac{1}{2}(7/3)^2=65/9\approx 7.222$	firm 1 **loses**
Firm 2, c_H	$(13/6)^2\approx 4.694$	$2^2=4$	high-cost type **gains**
Firm 2, c_L	$(19/6)^2\approx 10.028$	$(10/3)^2\approx 11.111$	low-cost type **loses**

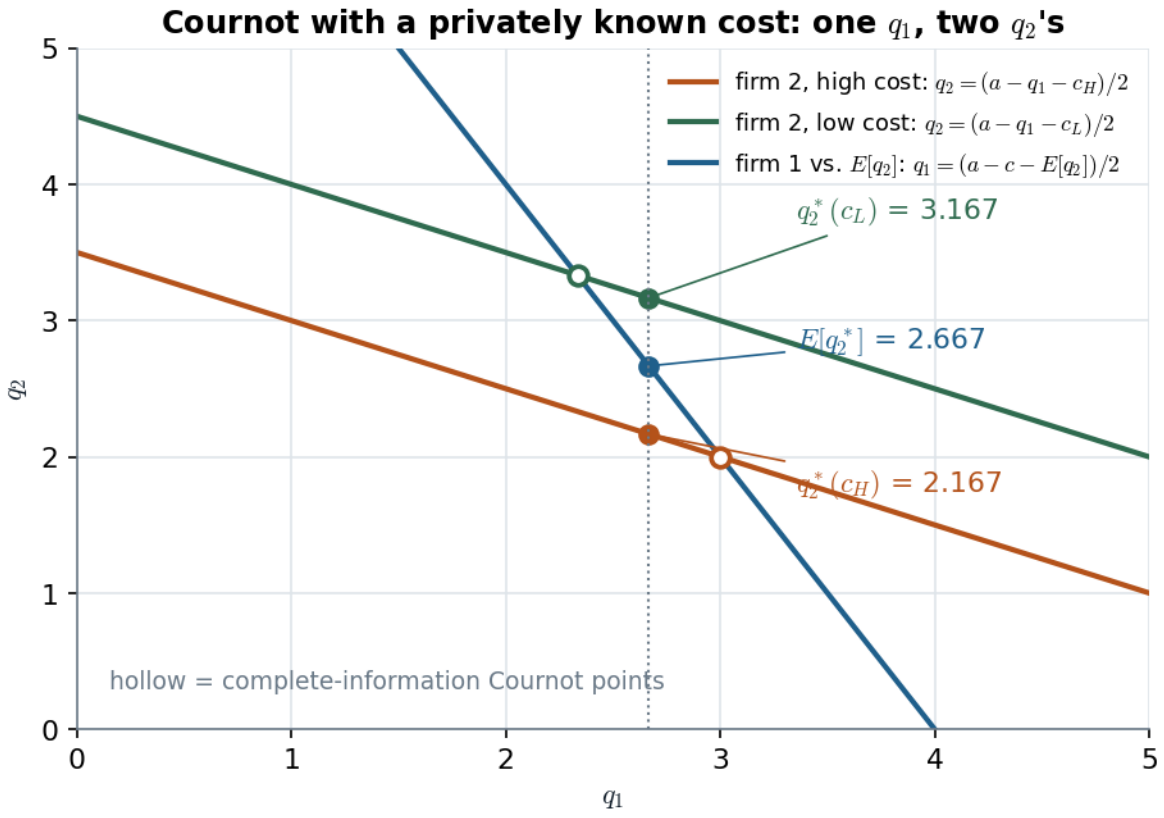


Figure — one q_1^*, two q_2^*'s: the filled points sit on the vertical $q_1 = 8/3$, the hollow complete-information points sit where firm 1's line would cross each of firm 2's lines if firm 1 knew the cost. Source: Gibbons pp. 145–146; calibration $a = 10$, $c = 2$, $c_H = 3$, $c_L = 1$, $\theta = \frac{1}{2}$ illustrative.

Firm 1's line is drawn against $E[q_2]$, so it crosses the vertical only at the mean 8/3. The high-cost point lies to the **left** of its complete-information twin (firm 1 produces less than the 3 it would against a known high-cost rival) and therefore **higher** on firm 2's high-cost line.

Links

- Prev: the BNE definition → L07-02-bayesian-nash-equilibrium.
- Complete-information Cournot (and its IESDS solution): Lecture 1, Lecture 2.
- Code: L07-12-python §3 (`cournot_bne`, checked against best-response iteration).
- **Exercises (number + page only):** Tadelis 12.2 (p. 266); Gibbons 3.2 and 3.3 (p. 169); MWG 8.E.3 (p. 265); Muñoz-Garcia Ch. 7 Ex. 5 (p. 336).

12.2.2 · 12.5 · Continuous types: best responses are thresholds, and mixed strategies are pure strategies of a perturbed game

Claim. With a continuum of types and two actions, a player's best response to *any* strategy of the other is a **cutoff in his own type** — so the search for a BNE collapses from functions to numbers. Two payoffs: the study-group game has the **unique** BNE threshold $c^{1/3}$; and **Harsanyi's purification** reads a mixed-strategy equilibrium as the limit of pure, type-contingent play when each player's payoff carries a small private perturbation.

Prev: L07-03-cournot-private-costs · Next: L07-05-adverse-selection-lemons · Map: L07-00-map

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1 · C — study groups (§12.2.2, pp. 255–258)

1.1 · The game

Two students hand in a joint lab. Each chooses effort $e_i = 1$ at cost $c < 1$, or shirks ($e_i = 0$, free). The lab succeeds if **at least one** works. Student i's type $\theta_i \sim U[0,1]$, independent; his value of success is θ_i^2.

i's choice	i's payoff
work, $e_i = 1$	$\theta_i^2 - c$ (success guaranteed)
shirk, $e_i = 0$	θ_i^2 if j works, 0 if j shirks

A pure strategy is a function $s_i : [0,1] \to \{0,1\}$ — a continuum of strategies, so neither a tree nor a matrix helps (p. 255).

1.2 · Only one number about j matters

Whatever $s_j(\cdot)$ is, i's payoff from shirking depends on it only through $\Pr\{s_j(\theta_j) = 1\}$. Working is a (weak) best response iff

$$\theta_i^2 - c \;\geq\; \theta_i^2 \Pr\{s_j(\theta_j) = 1\} \iff \theta_i^2(1 - \Pr\{s_j = 1\}) \geq c \iff \theta_i \;\geq\; \sqrt{\frac{c}{1 - \Pr\{s_j(\theta_j) = 1\}}}. \quad (12.3)$$

Claim 12.1 (p. 256). The best response of player i to any strategy $s_j(\cdot)$ is a **threshold rule**: there is $\hat{\theta}_i$ such that i works iff $\theta_i \geq \hat{\theta}_i$.

The right side of (12.3) is a constant for a given s_j, so the set of types who work is an upper interval. Intuition (p. 256): if j surely shirks, i works iff $\theta_i \geq \sqrt{c}$; any chance that j works lets i free-ride, so types just above $\sqrt{c}$ now shirk; if j works with high enough probability the cutoff exceeds 1 and i always shirks.

1.3 · Best response to a threshold

If j uses cutoff $\hat{\theta}_j$, then $\Pr\{s_j = 1\} = \Pr\{\theta_j \geq \hat{\theta}_j\} = 1 - \hat{\theta}_j$, and (12.3) becomes $\theta_i \geq \sqrt{c/\hat{\theta}_j}$ (12.4). A cutoff cannot exceed 1, so

$$BR_i(\hat{\theta}_j) = \begin{cases} \sqrt{c/\hat{\theta}_j} & \text{if } \hat{\theta}_j \geq c \\ 1 & \text{if } \hat{\theta}_j < c. \end{cases} \tag{12.5}$$

1.4 · The equilibrium: Claim 12.2 (p. 257)

Claim 12.2. In the unique BNE each player uses the same threshold $\hat{\theta}^*$, $0 < \hat{\theta}^* < 1$.

Interior candidates. Both on the top branch of (12.5):

$$\hat{\theta}_1 = \sqrt{c/\hat{\theta}_2}, \qquad \hat{\theta}_2 = \sqrt{c/\hat{\theta}_1} \tag{12.6}$$

Square both: $\hat{\theta}_1^2\hat{\theta}_2 = c$ and $\hat{\theta}_2^2\hat{\theta}_1 = c$. Divide the first by the second: $\hat{\theta}_1/\hat{\theta}_2 = 1$, so $\hat{\theta}_1 = \hat{\theta}_2 = \hat{\theta}^*$ and $(\hat{\theta}^*)^3 = c$:

$$\boxed{\hat{\theta}^* = c^{1/3}} \qquad (c < 1 \Rightarrow c < c^{1/2} < c^{1/3} < 1).$$

No corner equilibrium. Suppose $\hat{\theta}_j < c$. Then $\hat{\theta}_i = BR_i(\hat{\theta}_j) = 1$ (i never works). But then $BR_j(1) = \sqrt{c} > c$, contradicting $\hat{\theta}_j < c$. So the BNE strategies are

$$s_i^*(\theta_i) = \begin{cases} 0 & \theta_i < c^{1/3} \\ 1 & \theta_i \geq c^{1/3} \end{cases} \qquad i = 1, 2.$$

(The tie at $\theta_i = c^{1/3}$ is broken toward work; it has probability zero, footnote 4, p. 256.)

1.5 · Worked — $c = 1/8$ (illustrative)

Quantity	Value
Cutoff if the partner surely shirks, $\sqrt{c}$	0.354
BNE cutoff $c^{1/3}$	0.5
Check $(\hat{\theta}^*)^2\hat{\theta}^*$	$0.5^3 = 0.125 = c$ ✓
Probability a given student works, $1 - c^{1/3}$	0.5
Probability the lab fails (both shirk), $(c^{1/3})^2$	0.25

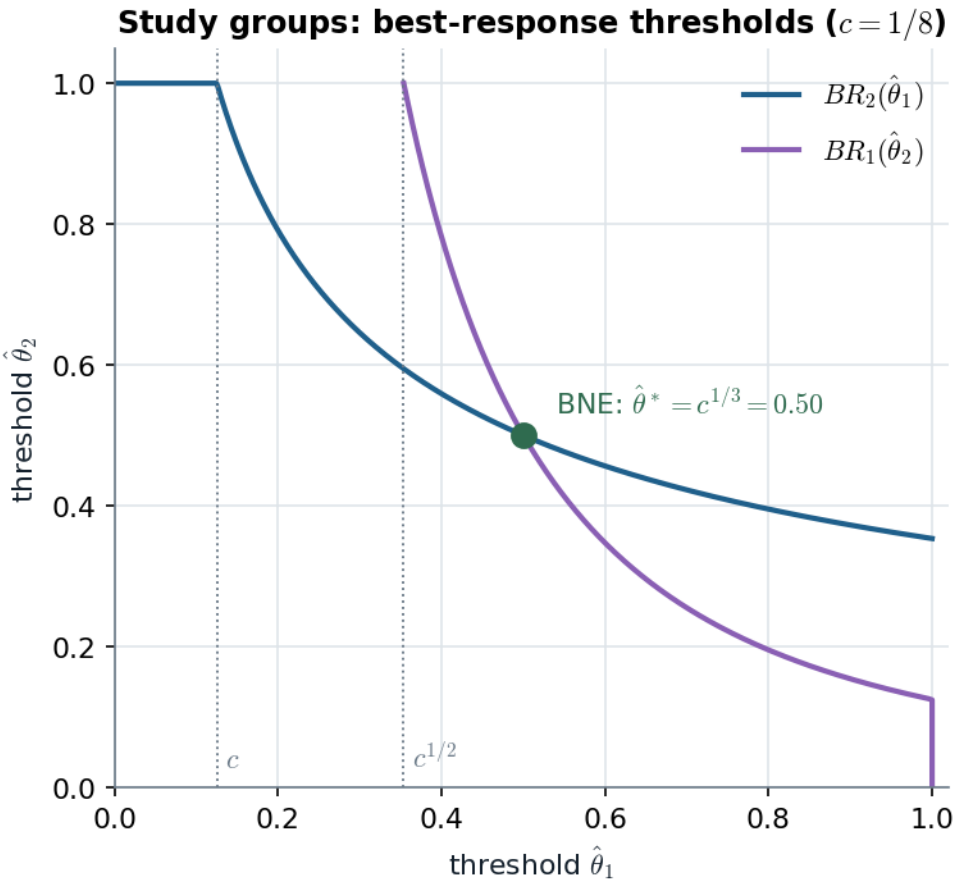


Figure — the two best-response threshold curves (12.5) cross once, at $c^{1/3}$, as in Tadelis's Fig. 12.4 (p. 257). $c = 1/8$ is illustrative; the book keeps c general.

Each curve is decreasing — **free-riding**: the harder my partner works (the lower his cutoff), the higher my own cutoff. The flat segment at 1 is the region $\hat{\theta}_j < c$, where j works so often that no type of i bothers. The curves cannot cross on that segment, which is the corner argument of §1.4 drawn.

MWG's identical example. MWG Example 8.E.2 (p. 256), the "Zigger" R&D consortium: types $U[0,1]$, benefit θ_i^2, cost c; condition (8.E.3) is (12.3) in other letters.

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2 · U — what the threshold result buys

- **From functions to numbers.** A strategy is any map $[0,1] \to \{0,1\}$; Claim 12.1 says only monotone cutoff maps can be best responses, so equilibrium is a fixed point in one number per player. The same reduction drives the auction notes: a bid function is pinned down by one differential equation (L07-07-first-price-dutch).
- **Monotonicity comes from the payoff, not from an assumption here**: the benefit of working, $\theta_i^2(1 - \Pr\{s_j = 1\})$, rises with θ_i while the cost c does not.

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3 · C — Harsanyi's interpretation of mixed strategies (§12.5, pp. 264–266)

3.1 · The discomfort

In Matching Pennies the unique equilibrium has each player mix 50–50, although each is **indifferent** between H and T at that point (Lecture 2). Why randomize in exactly that proportion?

3.2 · Perturbed Matching Pennies (p. 265)

	H	T
H	$1+\varepsilon_1,\ -1+\varepsilon_2$	$-1+\varepsilon_1,\ 1$
T	$-1,\ 1+\varepsilon_2$	$1,\ -1$

$\varepsilon_i \sim U[-\varepsilon, \varepsilon]$ independent, known only to i: a small private taste for heads. A strategy maps ε_i to $\{H, T\}$.

> **Claim 12.3.** The unique pure-strategy BNE is $s_i(\varepsilon_i) = H$ iff $\varepsilon_i \geq 0$ (and T otherwise). Its outcomes and payoffs converge to the mixed equilibrium of Matching Pennies as $\varepsilon \to 0$.

Verification of the equilibrium (p. 265): if j uses this rule, j plays H with probability $\Pr\{\varepsilon_j \geq 0\} = \frac{1}{2}$; facing a 50–50 opponent, i strictly prefers H iff $\varepsilon_i > 0$ — exactly the rule. Uniqueness takes more work (exercise 12.10, p. 269).

3.3 · Worked — Gibbons's Battle of the Sexes (§3.2.A, pp. 152–154)

Chris \ Pat	Opera	Fight
Opera	$2+t_c,\ 1$	$0, 0$
Fight	$0, 0$	$1,\ 2+t_p$

$t_c, t_p \sim U[0, x]$, independent, private. The complete-information game ($x = 0$) has a mixed equilibrium with Chris playing Opera and Pat playing Fight each with probability 2/3.

Candidate. Chris plays Opera iff $t_c \geq c$; Pat plays Fight iff $t_p \geq p$. Then Chris plays Opera with probability $(x-c)/x$, Pat plays Opera with probability p/x.

Chris's best response. Opera pays $\frac{p}{x}(2+t_c)$; Fight pays $(1-\frac{p}{x})\cdot 1$. Opera iff

$$\frac{p}{x}(2+t_c) \geq 1-\frac{p}{x} \iff 2+t_c \geq \frac{x}{p}-1 \iff t_c \geq \frac{x}{p}-3 = c. \tag{3.2.1}$$

Pat's best response. Fight pays $\frac{c}{x}(2+t_p)$; Opera pays $(1-\frac{c}{x})\cdot 1$. Fight iff

$$t_p \geq \frac{x}{c}-3 = p. \tag{3.2.2}$$

Solve. The two conditions are symmetric, so $p = c$ and $c = x/c - 3$, i.e. $c^2 + 3c - x = 0$:

$$c = \frac{-3+\sqrt{9+4x}}{2}, \qquad \Pr\{\text{Chris plays Opera}\} = \frac{x-c}{x} = 1 - \frac{-3+\sqrt{9+4x}}{2x}.$$

The limit. $\sqrt{9+4x} = 3\sqrt{1+4x/9} = 3 + \frac{2x}{3} + O(x^2)$, so $\frac{-3+\sqrt{9+4x}}{2x} \to \frac{1}{3}$ and the probability $\to \frac{2}{3}$.

x	1	0.5	0.1	0.01	$\to 0$
cutoff c	0.3028	0.1583	0.0330	0.0033	0
$\Pr\{\text{Opera}\} = 1 - c/x$	0.6972	0.6834	0.6703	0.6670	2/3

The point of purification (Gibbons p. 152): *"the crucial feature of a mixed-strategy Nash equilibrium is not that player j chooses a strategy randomly, but rather that player i is uncertain about player j's choice."* Each player is a pure-strategy type-contingent player; the population of types *looks* mixed. That is rules/game-theory item 8, now proved in an example.

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4 · G — the general theorem

Harsanyi's (1973) **purification theorem**: (almost) every mixed equilibrium of a complete-information game is the limit of pure-strategy BNE of nearby games with small independent payoff perturbations (Tadelis p. 265, which refers to Govindan 2003 for a short proof; Gibbons p. 152 notes the rare exceptions). Named, not examinable.

Links

- Prev: L07-03-cournot-private-costs · Next: L07-05-adverse-selection-lemons.
- Mixed strategies and indifference: Lecture 2.
- A threshold in a *buyer–seller* game: the lemons cutoff and the double auction's $v_b \geq v_s + \frac{1}{4} \to$ L07-05-adverse-selection-lemons.
- Code: L07-12-python §4 (`study_group_br`, `bos_purified`).
- **Exercises (number + page only):** Tadelis 12.9, 12.10 (p. 269), 12.7 (p. 268); Gibbons 3.5 (p. 170); Muñoz-Garcia Ch. 7 Ex. 6 (p. 339).

12.3 · Adverse selection: a single price buys the bottom of the pool, and trade unravels

Claim. When the seller knows the quality and the buyer does not, **any one price attracts exactly the sellers whose quality is below it**, so the buyer pays for an average that falls as the good types withdraw. In Tadelis's example every unit is worth 4 more to the buyer than to the seller, yet **only the lowest quality can trade in any BNE**. The same logic closes insurance markets (M&R, Cabral), and with **two-sided** private information no equilibrium of a double auction trades exactly when trade is efficient (Gibbons).

Prev: L07-04-threshold-strategies-purification · Next: L07-06-second-price-english · Map: L07-00-map

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1 · U — what perfect information was doing for markets (p. 258)

The competitive argument: if a good is held by someone who values it less than someone else, market pressure raises its price until the owner sells. With perfect information and no transaction costs, the **Coase theorem** says ownership ends up with whoever values the good most, whatever the initial allocation (Lecture 4 for bargaining). Akerlof (1970) asked what is left of this when the seller knows more than the buyer.

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2 · C — the orange grove (pp. 258–259)

Types. Player 1 owns a grove of quality $\theta_1 \in \{L, M, H\}$, each with probability $\frac{1}{3}$; only he knows it. **Values:**

Quality	Seller's value $v_1(\theta_1)$	Buyer's value $v_2(\theta_1)$	Gain from trade
L	10	14	4
M	20	24	4
H	30	34	4

Game (Fig. 12.5). Player 2 (a soybean grower) makes a take-it-or-leave-it offer $p \geq 0$; player 1 accepts (A) or rejects (R). A strategy for player 2 is a price; a pure strategy for player 1 is a map $s_1 : [0, \infty) \times \Theta_1 \to \{A, R\}$. Payoffs on acceptance: $p - v_1(\theta_1)$ to the seller, $v_2(\theta_1) - p$ to the buyer; zero otherwise.

Efficiency says every grove should change hands — the buyer values every quality more. **Under complete information it would:** with quality known to be L, the ultimatum-game SPE has the buyer offer 10 and the seller accept, and any price in $[10, 14]$ is a Nash equilibrium (p. 259, footnote 5).

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3 · C — unravelling, step by step (pp. 259–260)

Naive candidate. On average the land is worth $\frac{1}{3}(14{+}24{+}34) = 24$ to the buyer and $\frac{1}{3}(10{+}20{+}30) = 20$ to the seller. Offer $p = 20$?

Step	Price range	Who accepts	Buyer's value of the accepting pool	Buyer's payoff
1	$p \in [20, 30)$	L, M (each $\frac{1}{2}$ given a sale, footnote 6)	$\frac{1}{2} \cdot 14 + \frac{1}{2} \cdot 24 = 19$	$19 - p < 0$
2	$p \geq 30$	L, M, H	24	$24 - p \leq -6 < 0$
3	$p \in [10, 20)$	L only	14	$14 - p \geq 0$ iff $p \leq 14$
4	$p < 10$	nobody	—	0

The Bayes step in row 1: conditional on a sale, the grove is L or M with prior weights $\frac{1}{3}, \frac{1}{3}$, renormalized to $\frac{1}{2}, \frac{1}{2}$.

> **Proposition 12.1** (p. 260). Trade can occur in a Bayesian Nash equilibrium only if it involves the lowest type of player 1 trading. Furthermore, any price $p^* \in [10, 14]$ can be supported as a Bayesian Nash equilibrium.

The supporting strategies (p. 260): player 2 offers p^*; player 1 of type L accepts iff $p \geq p^*$, type M iff $p \geq 20$, type H iff $p \geq 30$. **Check player 2:** at p^* he gets $\frac{1}{3}(14 - p^*) \geq 0$; a lower price is refused (0); a price in $(p^*, 20)$ buys the same L for more; $[20, 30)$ gives $\frac{2}{3}(19 - p) < 0$; $p \geq 30$ gives $24 - p < 0$. So p^* is a best response iff $p^* \leq 14$. **Check player 1:** at $p^* \in [10, 14]$, type L gains $p^* - 10 \geq 0$ by accepting, and M, H would lose by accepting.

> **Adverse selection** (p. 260): when the buyer offers his *average* value, the sellers willing to sell at that price are *below* average — the best types select themselves out — and in the example the traded quality falls all the way to the bottom.

Two things make it bite (pp. 260–261):

1. **Common values.** The seller's type enters the **buyer's** payoff (v_2 depends on θ_1). With private values the buyer would not care who accepts.
2. **Talk is cheap.** "My soil is of high quality" is worthless: every type would say it. A **credible** revelation of type changes the outcome — that is signaling, Lecture 8 (Tadelis Ch. 16).

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4 · Worked — how much premium does trade need?

Generalize the grove: K equally likely qualities with seller values $v_k = v_1 + (k-1)s$ and buyer values $v_k + \Delta$. Qualities $1, \ldots, k$ can trade iff the buyer does not lose on that pool at the price v_k that brings in the k-th:

$$\underbrace{v_1 + \tfrac{(k-1)s}{2}}_{\text{mean seller value of } 1..k} + \Delta \;\geq\; \underbrace{v_1 + (k-1)s}_{v_k} \iff \boxed{\Delta \geq \frac{(k-1)s}{2}}$$

Tadelis ($s = 10$, $K = 3$)	$k = 1$ (L)	$k = 2$ (L, M)	$k = 3$ (all)
premium needed	0	5	10
at $\Delta = 4$	trades	out	out

- **The level v_1 drops out**: only the **spread of quality the seller knows about**, relative to the gain from trade, matters.
- **Gains lost.** Full information: every grove trades, expected gain $\Delta = 4$. Best BNE: only L, expected gain $\frac{1}{3} \cdot 4 = \frac{4}{3}$ — **two thirds of the surplus destroyed**.
- **The buyer's preferred equilibrium.** Offering v_k yields him $\frac{k}{K}(\Delta - \frac{(k-1)s}{2})$; at $\Delta = 4$ only $k = 1$ is positive: offer 10, expected surplus $\frac{1}{3} \cdot 4 = \frac{4}{3}$.

  Interactive companion. The Lemons Ladder (`companions/companion-lemons.html`) — move the premium Δ and the spread s and watch the highest tradable quality jump at $\Delta = s(k-1)/2$; move v_1 and see the prices shift while the trade set stays put.

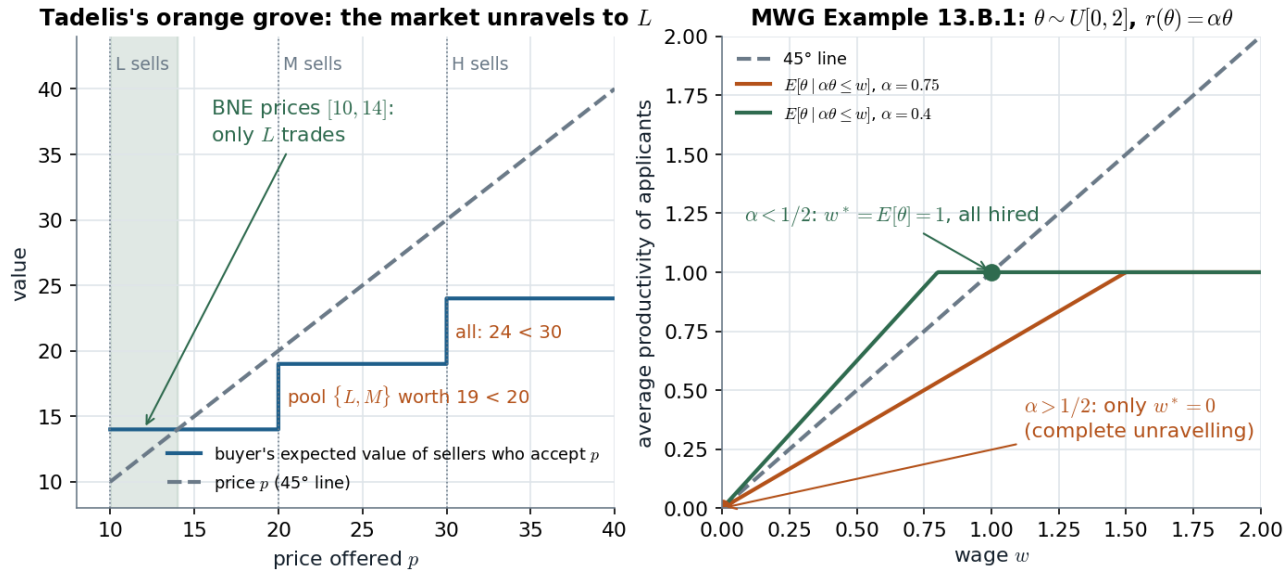


Figure — trade is possible where what a price buys (the step) is at least the price (the 45° line). Left: Tadelis pp. 258–260, the pools are worth 14, 19, 24 at prices 10, 20, 30. Right: MWG Example 13.B.1, p. 442, with $\alpha = 0.75$ and 0.4 illustrative.

Every jump of the step comes **too late**: when the price reaches 20 the pool improves to 19, one short of the price that bought the improvement; at 30 it improves to 24, six short. The continuous version on the right is the same picture with a continuum of qualities (§8).

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5 · C — insurance: the buyer is the informed side (M&R pp. 149–152)

Why insurance. The people who buy a policy are not a random sample: they know something about their expected claims (M&R's maternity-coverage example, p. 149). Here the **buyer** has the private information, and the bad types are the **high-cost** ones.

M&R's model. Expected claims $x \sim U[0, \bar{x}]$, private. A buyer also values the risk reduction at v, so she buys at price P iff $P \le v + x$, i.e. iff $x \ge P - v$. The insurer pays $1 + c$ per dollar of claims (c = administration cost). If the marginal buyer is x, the pool's average claim is $(x + \bar{x})/2$:

$$P_s(x) = \tfrac{1}{2}(x + \bar{x})(1 + c) \quad \text{(insurer breaks even)} \tag{5.1}$$

$$P_b(x) = x + v \quad \text{(buyer } x \text{ is just willing)} \tag{5.2}$$

When does a market exist? Need some x with $P_b(x) \geq P_s(x)$. The gap

$$P_b(x) - P_s(x) = \frac{1-c}{2}x + v - \frac{1+c}{2}\bar{x}$$

rises with x (for $c < 1$), so it is largest at $x = \bar{x}$, where it equals $v - c\bar{x}$:

$$\boxed{\text{market breaks down} \iff c\,\bar{x} > v} \qquad \text{(p. 152)}$$

The highest-claim customer must be worth serving on her own. **Mandatory coverage** (everyone in) has average benefit $v + \bar{x}/2$ and average cost $(1+c)\bar{x}/2$: worthwhile iff $v > c\bar{x}/2$. Hence the band

$$c\,\bar{x} > v > \tfrac{c\,\bar{x}}{2}: \quad \text{insurance is desirable, but no private individual market survives}$$

— the case for **group** (employer) insurance, a private-sector fix (p. 152).

Illustration ($\bar{x} = 100$, $c = 0.1$, illustrative): the market exists iff $v \geq 10$; mandatory coverage pays iff $v > 5$; at $v = 8$ insurance is efficient and the market is closed.

Credit rationing (M&R p. 153). Safe borrowers A repay 1.1M for sure; risky B's projects pay 0.9M or 1.3M. Raising the loan rate to just over 10% drives out A (they would pay more than they earn) and keeps B (expected profit $\frac{1}{2}(1.3 - 1.1) = 0.1$M), so the bank's pool worsens and its return falls. **The price the lender charges changes the quality of its customers** — Stiglitz–Weiss rationing.

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6 · Worked — Cabral's insurer (§7.4, pp. 174–176)

10% of patients are high risk. Values of a year's insurance: 20 (high), 3 (low); costs to the insurer: 30 (high), 1 (low), in \$000. Averages: value $0.1 \cdot 20 + 0.9 \cdot 3 = 4.7$, cost $0.1 \cdot 30 + 0.9 \cdot 1 = 3.9$. The insurer considers $p = 4.5$ (between the averages) and $p = 30$.

Price	High risk ($\frac{1}{10}$)	Low risk ($\frac{9}{10}$)	Insurer's expected profit
4.5	buys ($20 \geq 4.5$): insurer $4.5 - 30 = -25.5$	does not buy ($3 < 4.5$)	$0.1(-25.5) = -2.55$
30	does not buy	does not buy	0

The insurer sets $p = 30$ and **nobody is insured** (Cabral prints −2.5, rounding −2.55). Reverse the information — the patient does not know his own risk either (Cabral's Fig. 7.12) — and $p = 4.5$ is accepted by the "average" patient ($4.7 \geq 4.5$), earning $4.5 - 3.9 = 0.6 > 0$. **Same numbers; only who knows what, and when, changed** (p. 176).

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7 · C — two-sided private information: the double auction (Gibbons §3.2.C, pp. 158–163)

7.1 · The game

A seller (value v_s) and a buyer (value v_b), **each privately informed**, $v_s, v_b \sim U[0,1]$ independent. Simultaneously the seller names an ask p_s and the buyer an offer p_b. If $p_b \geq p_s$ they trade at $p = (p_b + p_s)/2$; otherwise no trade. Payoffs: buyer $v_b - p$, seller $p - v_s$ on trade, 0 otherwise. Strategies are functions $p_b(v_b)$, $p_s(v_s)$. Efficient trade: iff $v_b \geq v_s$.

7.2 · Many equilibria: the one-price family (p. 159–160)

For any $x \in [0,1]$: the buyer offers x if $v_b \geq x$ (else 0); the seller asks x if $v_s \leq x$ (else 1). Given the other's strategy, each side's choice is just "trade at x or not", and exactly the types who prefer trading at x do so. **Trade iff $v_s \leq x \leq v_b$.** Expected gains:

$$\int_0^x \int_x^1 (v_b - v_s)\, dv_b\, dv_s = x(1-x)\Big[\tfrac{1+x}{2} - \tfrac{x}{2}\Big] = \frac{x(1-x)}{2}, \quad \max \tfrac{1}{8} \text{ at } x = \tfrac{1}{2}.$$

7.3 · The linear equilibrium, derived (pp. 160–161)

Buyer's best reply to a linear seller, $p_s = a_s + c_s v_s$, so $p_s \sim U[a_s, a_s + c_s]$: given trade, the expected ask is $E[p_s \mid p_s \leq p_b] = (a_s + p_b)/2$ and $\Pr\{p_s \leq p_b\} = (p_b - a_s)/c_s$. The buyer solves

$$\max_{p_b} \Big[v_b - \tfrac{1}{2}\Big(p_b + \tfrac{a_s + p_b}{2}\Big)\Big] \frac{p_b - a_s}{c_s} = \Big[v_b - \frac{3p_b + a_s}{4}\Big] \frac{p_b - a_s}{c_s}.$$

FOC: $-\frac{3}{4}(p_b - a_s) + v_b - \frac{3p_b + a_s}{4} = 0 \Rightarrow 4v_b = 6p_b - 2a_s$:

$$p_b = \tfrac{2}{3} v_b + \tfrac{1}{3} a_s. \tag{3.2.5}$$

Seller's best reply to a linear buyer, $p_b = a_b + c_b v_b \sim U[a_b, a_b + c_b]$; write $A = a_b + c_b$:

$$\max_{p_s} \Big[\tfrac{1}{2}\Big(p_s + \tfrac{p_s + A}{2}\Big) - v_s\Big] \frac{A - p_s}{c_b} \;\Rightarrow\; \tfrac{3}{4}(A - p_s) - \Big(\tfrac{3p_s + A}{4} - v_s\Big) = 0 \;\Rightarrow\; p_s = \tfrac{2}{3} v_s + \tfrac{1}{3}(a_b + c_b). \tag{3.2.6}$$

Fixed point. (3.2.5): $c_b = \frac{2}{3}$, $a_b = a_s/3$. (3.2.6): $c_s = \frac{2}{3}$, $a_s = (a_b + c_b)/3$. Then $a_s = \frac{1}{3}(\frac{a_s}{3} + \frac{2}{3}) \Rightarrow \frac{8}{9} a_s = \frac{2}{9} \Rightarrow a_s = \frac{1}{4}$, $a_b = \frac{1}{12}$:

$$\boxed{p_b(v_b) = \tfrac{2}{3} v_b + \tfrac{1}{12}, \qquad p_s(v_s) = \tfrac{2}{3} v_s + \tfrac{1}{4}} \tag{3.2.7–3.2.8}$$

When do they trade? $p_b \geq p_s \iff \frac{2}{3}(v_b - v_s) \geq \frac{1}{4} - \frac{1}{12} = \frac{1}{6} \iff$

$$\boxed{v_b \geq v_s + \tfrac{1}{4}}$$

Seller types above $\frac{3}{4}$ ask more than the highest offer $p_b(1) = \frac{3}{4}$; buyer types below $\frac{1}{4}$ offer less than the lowest ask $p_s(0) = \frac{1}{4}$ (p. 161).

7.4 · How much is lost

Equilibrium	Trade iff	Expected gains	Share of efficient
Efficient benchmark	$v_b \geq v_s$	$\int_0^1 \int_{v_s}^1 (v_b - v_s) = \frac{1}{6} \approx 0.1667$	100%
Linear	$v_b \geq v_s + \frac{1}{4}$	$\frac{9}{64} \approx 0.1406$	84.4%
One-price, $x = \frac{1}{2}$	$v_s \leq \frac{1}{2} \leq v_b$	$\frac{1}{8} = 0.125$	75%

The linear figure: for fixed $v_s \leq \frac{3}{4}$, $\int_{v_s + 1/4}^1 (v_b - v_s)\, dv_b = \frac{1}{2}[(1 - v_s)^2 - \frac{1}{16}]$; with $u = 1 - v_s$, $\int_{1/4}^1 \frac{1}{2}(u^2 - \frac{1}{16})\, du = [\frac{u^3}{6} - \frac{u}{32}]_{1/4}^1 = \frac{52}{384} + \frac{2}{384} = \frac{9}{64}$.

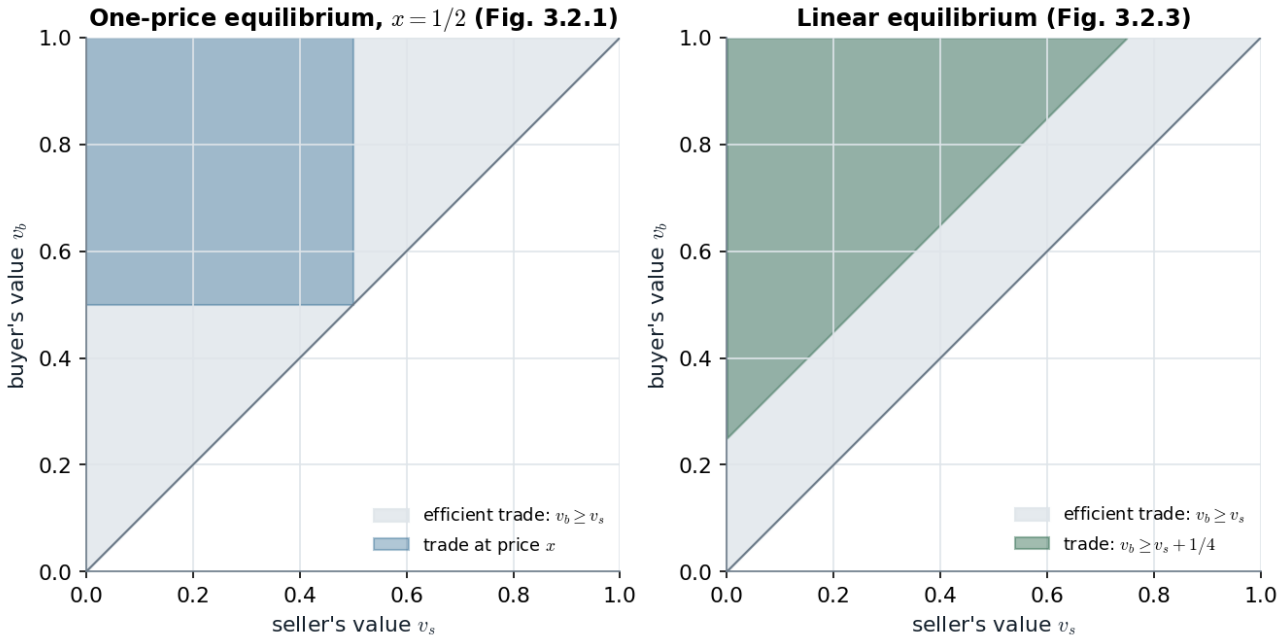


Figure — trade regions inside the efficient triangle $v_b \geq v_s$: the one-price equilibrium misses valuable trades and makes some worth almost nothing; the linear one misses only trades worth less than $\frac{1}{4}$. Source: Gibbons Figs. 3.2.1 and 3.2.3, pp. 160–163.

> **Myerson–Satterthwaite (named only).** Among all BNE of the double auction, the linear one gives the highest expected gains; and **no** bargaining game the two would willingly play trades if and only if $v_b \geq v_s$ (Gibbons pp. 162–163). Proving it uses the revelation principle — mechanism design, outside this course's scope.

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8 · G — the competitive version (MWG §13.B, pp. 437–447)

Workers of productivity θ (private) earn $r(\theta)$ at home; firms pay one wage w to all.

- **Definition 13.B.1** (p. 439): a competitive equilibrium is (w^*, Θ^*) with $\Theta^* = \{\theta : r(\theta) \leq w^*\}$ and $w^* = E[\theta \mid \theta \in \Theta^*]$; hence (13.B.6), p. 441: $w^* = E[\theta \mid r(\theta) \leq w^*]$.
- **Example 13.B.1** (p. 442): $\theta \sim U[0,2]$, $r(\theta) = \alpha\theta$, $\alpha < 1$ (every worker belongs in a firm). Applicants at wage w are $\theta \leq w/\alpha$, average $w/(2\alpha)$ while $w \leq 2\alpha$. For $\alpha > \frac{1}{2}$, $w/(2\alpha) < w$ for all $w > 0$: **the only equilibrium is** $w^* = 0$ — complete unravelling (figure above, right panel, $\alpha = 0.75$).
- Multiple equilibria can be Pareto-ranked (Fig. 13.B.3, p. 442); **Prop. 13.B.1** (p. 443): with wage-setting firms, the SPNE picks the highest-wage competitive equilibrium; **Prop. 13.B.2** (p. 447): that equilibrium is a **constrained** Pareto optimum — a planner who cannot see θ cannot improve on it. Named, not examinable.

Links

- Prev: L07-04-threshold-strategies-purification · Next: auctions, L07-06-second-price-english.
- The same conditioning in auctions and votes: L07-09-common-values-curses.
- Signaling and screening as the market's answer: Lecture 8.

- Rules pack: Topic 7 (adverse selection; its lemons section is still a placeholder that this note can fill).
- Code: L07-12-python §5–6 (`lemons_trade`, `mr_market_exists`, `insurer_payoff`, `mwg_wages`, `linear_double_auction`).
- **Exercises (number + page only):** Tadelis 12.5, 12.6 (pp. 267–268); Gibbons 3.8 (p. 171); MWG 13.B.1–13.B.5 (p. 473); M&R Questions 5.3–5.5 (pp. 164–165); Cabral 7.9 (p. 180), 7.10 (p. 181).

13.1.1–13.1.2 · Second-price and English auctions: bid your value, whatever the others do

> **Claim.** In the second-price sealed-bid auction **bidding one's value is weakly dominant** — a Bayesian Nash equilibrium that needs no beliefs at all — and the English (button) auction has the same outcome: the highest-value bidder wins and pays the second-highest value. Lecture 1 met the dominance argument as a complete-information game; here it becomes a **strategy** $s_i(\theta_i) = \theta_i$, a map from types to bids.

Prev: L07-05-adverse-selection-lemons · Next: L07-07-first-price-dutch · Map: L07-00-map

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1 · C — four auctions, one Bayesian game (pp. 270–272)

	Open (price evolves in view)	**Sealed bid** (simultaneous)
Pay the second-highest price	**English**: price rises until one bidder is left; he pays where the last rival quit	**Second-price (Vickrey)**: highest bid wins, pays the highest losing bid
Pay your own price	**Dutch**: price falls until someone shouts "buy" and pays that price	**First-price**: highest bid wins, pays his bid

The table's rows are the two strategic equivalences this note and the next establish (English ≈ second-price; Dutch ≡ first-price). **An auction is a Bayesian game** (p. 271): players are bidders, actions are bids, payoffs depend on winning and paying, and nobody knows the others' valuations.

The independent private values (IPV) setting (p. 272). Bidder i's value is his type $\theta_i \in [\underline{\theta}_i, \overline{\theta}_i]$, drawn from F_i, **independently** across bidders. Winning at price p pays $\theta_i - p$; losing pays 0. Private values: i's payoff depends on his own type only (the filet mignon that must be eaten within the hour, p. 272). The seller is not a player: once he fixes the rules, he has no further move.

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2 · C — the second-price sealed-bid auction (pp. 272–274)

Payoff (p. 273), with $b_j^* \equiv \max_{j \neq i} b_j$:

$$u_i(b_i, b_{-i}; \theta_i) = \begin{cases} \theta_i - b_j^* & \text{if } b_i > b_j \ \forall j \neq i \\ 0 & \text{if } b_i \leq b_j \text{ for some } j \neq i \end{cases}$$

(Tadelis's baseline tie rule: tied top bidders get nothing; the Remark on p. 274 replaces it with a random winner paying his own bid, and nothing below changes.)

Strategy: a function $s_i : [\underline{\theta}_i, \overline{\theta}_i] \to \mathbb{R}_+$.

> **Proposition 13.1** (p. 273). In the second-price sealed-bid auction, each player has a weakly dominant strategy, which is to bid his true valuation: $s_i(\theta_i) = \theta_i$ for all $i \in I$ is a Bayesian Nash equilibrium in weakly dominant strategies.

2.1 · Proof, the three cases (pp. 273–274, Fig. 13.1)

Fix θ_i and an underbid $b_i < \theta_i$. Let b_j^* be the highest rival bid. Split the line into region L ($b_j^* < b_i$), region M ($b_i < b_j^* < \theta_i$), region H ($b_j^* > \theta_i$).

Case	Highest rival bid	Bid $b_i < \theta_i$	Bid θ_i	Verdict
1	region L	wins, pays b_j^*	wins, pays b_j^* — **same price**	equal
2	region H	loses, 0	loses, 0	equal
3	region M	loses, 0	wins, $\theta_i - b_j^* > 0$	**truthful strictly better**

Never worse, sometimes strictly better: θ_i weakly dominates every lower bid. An overbid $b_i > \theta_i$ is the mirror image — it changes the outcome only when $\theta_i < b_j^* < b_i$, where it wins at a price above the value (Tadelis leaves the write-up as exercise 13.1, p. 285).

> **Why it works: your bid decides *whether* you win, never *what you pay*.** The price is set by someone else's bid, so the only question your bid answers is "do I want the object at the price the others set?" — and bidding θ_i answers "yes" exactly when θ_i exceeds that price.



Figure — the three regions of Tadelis's Fig. 13.1 as payoff curves: underbidding forfeits region M, overbidding buys losses between θ_i and b_i, elsewhere all bids tie. Source: Prop. 13.1, pp. 273–274; $\theta_i = 0.6$, $b_i = 0.4$, $b_i' = 0.8$ illustrative.

The truthful curve is the upper envelope of the other two everywhere: that is weak dominance read off a picture. At $m = 0.5$ (region M) truthful earns 0.1 and the underbid earns 0.

2.2 · What the dominance buys (p. 274)

1. **No beliefs needed.** The bidder need not know F_j or the others' strategies: the common-prior assumption can be dropped.
2. **Robust to correlation.** With correlated but still *private* values, truth-telling stays weakly dominant.
3. **Efficiency.** The highest-value bidder wins: the outcome is Pareto optimal.

2.3 · Worked — three bidders

Values 0.9, 0.6, 0.3; everyone bids his value. Bidder 1 wins and pays 0.6; payoffs $(0.3, 0, 0)$; seller's revenue 0.6 = the **second-highest value**. Could bidder 2 do better by bidding 0.95? He would win and pay $0.9 > 0.6$: payoff -0.3. Could bidder 1 gain by bidding 0.7 or 2? He still wins at 0.6: no change.

Dominance is not uniqueness. Weak dominance singles out this equilibrium; it does not rule out others built on weakly dominated bids (whether they exist under complete information is Tadelis exercise 13.2, p. 285; Lecture 1 showed that IESDS alone says nothing in this auction).

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3 · C — the English auction as a button auction (pp. 275–276)

An open ascending auction has no best response with continuous increments (any jump could be smaller), as in Bertrand with unequal costs (Lecture 2). Milgrom and Weber's **button auction** fixes this: the price rises continuously from 0; each bidder holds a button and **quits for good** by releasing it; the last bidder holding wins and pays the price at which the second-to-last let go. A strategy maps a type to a quitting price.

> **Proposition 13.2** (p. 275). In the button-auction model it is a weakly dominant strategy for each player to keep his button pressed as long as $p < \theta_i$ and to release it once $p = \theta_i$. This is outcome-equivalent to the second-price sealed-bid auction.

Proof sketch (p. 276). While $p < \theta_i$, staying in can only lead to winning at a price below θ_i; quitting forgoes that. At $p = \theta_i$, staying could only win at a price above value. The winner is the highest-value bidder and pays the price at which the second-highest quit — **his value**. Same allocation, same price as §2.

	Second-price sealed	English (button)
Weakly dominant strategy	bid θ_i	quit at $p = \theta_i$
Winner	highest θ	highest θ
Price	second-highest θ	second-highest θ

eBay (p. 276) runs proxy bidding: your "maximum bid" is a button you hold for you, and eBay's own advice, "enter the maximum amount you are willing to pay", is the weakly dominant strategy.

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4 · U — why sell by auction at all (Cabral §6.4, pp. 140–142)

Two buyers whose values are **both** 100 or **both** 150, equally likely; the buyers know, the seller does not.

Selling method	Revenue
Posted price 100	100 for sure
Posted price 150	$\frac{1}{2} \cdot 150 + \frac{1}{2} \cdot 0 = 75$
Ascending auction	stops at 100 if values are 100, at 150 if 150: $\frac{1}{2} \cdot 100 + \frac{1}{2} \cdot 150 = 125$

An auction is price discrimination by self-selection: it makes the price track the buyers' valuations without the seller knowing them (p. 141). Cabral's bidding-strategy remark (p. 141) is the first-price trade-off of the next note: a higher bid wins more often but earns less when it wins, *"just like monopoly pricing"*.

Links

- Prev: L07-05-adverse-selection-lemons · Next: the format where you pay your bid, L07-07-first-price-dutch.
- **Lecture 1 already proved the dominance** with a reserve price on a bid grid: Lecture 1 · `Listas/lista-lecture-01.tex` Q1 b) and its check `Listas/lista-lecture-01-check.py::spa_payoff` · `Listas/pratica-lecture-01.md` items 17 (Tadelis 4.3, p. 76), 23 (MWG 8.B.3, p. 262) and 24, with solutions in `Listas/pratica-lecture-01-solutions/item23.tex`, `item24.tex` · quiz `Simulados/quiz-lecture01-2-dominance-pareto-2026-10-13.md` (section *auction*).
- Revenue of the format: L07-08-revenue-equivalence. With common values truth-telling fails: L07-09-common-values-curses.
- **Interactive companion.** The Bid Shading Desk (`companions/companion-auction-bidding.html`) — press "b = θ" and read the second-price payoff tile at its optimum; move b either way and the loss tile turns positive.
- Code: L07-12-python §7 (`spa_payoff`, the three cases).
- **Exercises (number + page only):** Tadelis 13.1, 13.2 (p. 285), 13.3 (pp. 285–286), 13.4 (p. 286); Muñoz-Garcia Ch. 8 Ex. 4 (p. 362).

13.1.3 · First-price and Dutch auctions: shade the bid until margin and win probability balance

Claim. When the winner pays his own bid, bidding one's value guarantees zero, so every bidder **shades**. With no dominant strategy, the bid depends on beliefs about the others' bids, and the symmetric BNE solves one differential equation: $s(\theta) = \theta - \int_{\underline{\theta}}^{\theta} F^{n-1}/F(\theta)^{n-1}$, which for uniform values is $\boxed{s(\theta) = \frac{n-1}{n}\theta}$. The Dutch auction is the same game in different clothes.

Prev: L07-06-second-price-english · Next: L07-08-revenue-equivalence · Map: L07-00-map

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1 · C — bidding your value is dominated (p. 276)

Claim 13.1. In a first-price sealed-bid auction it is a dominated strategy for a player to bid his valuation.

Bidding θ_i pays 0 if you lose and $\theta_i - \theta_i = 0$ if you win: **zero always**. Bidding slightly less wins with positive probability (against low enough rival bids) at a positive margin: expected payoff > 0. The domination is **weak** — against rival bids above θ_i both bids earn 0.

The trade-off (p. 276): lowering the bid raises the margin $\theta_i - b_i$ when you win (marginal benefit) and lowers the probability of winning (marginal cost). Unlike §13.1.1, the best bid now **depends on the others' strategies** — no weakly dominant strategy exists (p. 277).

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2 · C — the expected payoff (p. 277)

Assumption 13.1. The higher a player's valuation, the higher his bid: $\theta_j > \theta_j' \Rightarrow s_j(\theta_j) > s_j(\theta_j')$.

So s_j is invertible: a bid b comes from exactly one type $s_j^{-1}(b)$, and

$$\Pr\{s_j(\theta_j) < b_i\} = \Pr\{\theta_j < s_j^{-1}(b_i)\} = F_j(s_j^{-1}(b_i)).$$

Independence multiplies these across rivals:

$$E_{\theta_{-i}}\big[u_i(b_i, s_{-i}(\theta_{-i}); \theta_i) \mid \theta_i\big] = \prod_{j \neq i} F_j(s_j^{-1}(b_i)) \times (\theta_i - b_i). \tag{13.1}$$

Here the common prior has real bite (p. 277): F_j and s_j both enter (13.1). In the second-price auction neither did.

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3 · C — the symmetric equilibrium, every step (pp. 277–279)

Symmetric bidders: values on $[\underline{\theta}, \overline{\theta}]$ with common F, density f; all rivals use $s(\cdot)$. Type θ_i solves

$$\max_{b \geq 0} \ \big[F(s^{-1}(b))\big]^{n-1}(\theta_i - b). \tag{13.2}$$

FOC (interior), product rule:

$$-\big[F(s^{-1}(b))\big]^{n-1} + (n-1)\big[F(s^{-1}(b))\big]^{n-2} f(s^{-1}(b))\,\frac{d\,s^{-1}(b)}{db}\,(\theta_i - b) = 0. \tag{13.3}$$

Impose equilibrium. $b = s(\theta)$ must solve (13.3) for every θ. Then $s^{-1}(b) = \theta$ and $\frac{d\,s^{-1}(b)}{db} = \frac{1}{s'(\theta)}$:

$$-[F(\theta)]^{n-1} + (n-1)[F(\theta)]^{n-2} f(\theta)\,\frac{\theta - s(\theta)}{s'(\theta)} = 0. \tag{13.4}$$

Rearrange. Multiply by $s'(\theta)$ and move the $s(\theta)$ term left:

$$[F(\theta)]^{n-1} s'(\theta) + (n-1)[F(\theta)]^{n-2} f(\theta) s(\theta) = (n-1)[F(\theta)]^{n-2} f(\theta)\,\theta. \tag{13.5}$$

Spot the product rule. The left side is $\frac{d}{d\theta}([F(\theta)]^{n-1} s(\theta))$, since $\frac{d}{d\theta}[F]^{n-1} = (n-1)[F]^{n-2} f$. Integrate from $\underline{\theta}$ (where $F = 0$):

$$[F(\theta)]^{n-1} s(\theta) = \int_{\underline{\theta}}^{\theta} (n-1)[F(x)]^{n-2} f(x)\,x\,dx. \tag{13.6}$$

Integrate by parts (footnote 5, p. 278) with $g'(x) = (n-1)[F(x)]^{n-2} f(x)$, $g = [F]^{n-1}$, $h(x) = x$:

$$\int_{\underline{\theta}}^{\theta} (n-1)[F(x)]^{n-2} f(x)\,x\,dx = [F(\theta)]^{n-1}\theta - \int_{\underline{\theta}}^{\theta} [F(x)]^{n-1} dx$$

(the boundary term at $\underline{\theta}$ vanishes because $F(\underline{\theta}) = 0$). Divide by $[F(\theta)]^{n-1}$:

$$\boxed{s(\theta) = \theta - \frac{\int_{\underline{\theta}}^{\theta} [F(x)]^{n-1} dx}{[F(\theta)]^{n-1}}} \tag{13.7}$$

The bid is below the value, by an amount that depends on F, n and θ; this is the unique symmetric BNE (p. 278, proof beyond the book).

3.1 · Uniform values on $[0,1]$ (p. 279)

$F(x) = x$: $\int_0^{\theta} x^{n-1} dx = \theta^n/n$, so

$$s(\theta) = \theta - \frac{\theta^n/n}{\theta^{n-1}} = \theta - \frac{\theta}{n} = \boxed{\theta\,\frac{n-1}{n}} \tag{13.8}$$

The **bid-shading factor** $(n-1)/n \in (0,1)$ rises with n: more bidders, fiercer competition, bids closer to values.

n	2	3	5	10	$\to \infty$
$s(\theta)/\theta$	0.5	0.667	0.8	0.9	1

3.2 · A reading of (13.7) that pays off in the next note

The highest of the $n-1$ rival values, Y, has c.d.f. F^{n-1}. Integrating by parts as above, $E[Y \mid Y < \theta] = \theta - \int_{\underline{\theta}}^{\theta} F^{n-1}/F(\theta)^{n-1}$. So

> **the first-price bid equals the expected highest rival value, given that it is below yours** — exactly the price you would expect to pay *when you win* in a second-price auction. Uniform case: $E[Y \mid Y < \theta] = \frac{n-1}{n}\theta$. Revenue equivalence (L07-08-revenue-equivalence) is this line averaged over types.

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4 · Worked — Gibbons's two-bidder auction (pp. 155–158)

Two bidders, $v_i \sim U[0,1]$ independent, highest bid wins and pays it, ties by coin flip.

Guess linear, $b_j(v_j) = a_j + c_j v_j$ (not a restriction on strategies — a question whether a linear equilibrium exists). For $a_j \le b_i \le a_j + c_j$:

$$\max_{b_i} (v_i - b_i)\Pr\{b_i > a_j + c_j v_j\} = (v_i - b_i)\frac{b_i - a_j}{c_j}.$$

FOC: $-(b_i - a_j) + (v_i - b_i) = 0 \Rightarrow b_i = \dfrac{v_i + a_j}{2}$ when $v_i \ge a_j$ (else bid a_j, lose). Linearity rules out $0 < a_j < 1$ (the reply would be flat, then rising) and $a_j \ge 1$ (it would mean $b_j(v_j) \ge v_j$), so $a_j \le 0$, giving $c_i = \frac{1}{2}$, $a_i = a_j/2$; symmetry gives $a = a/2 = 0$:

$$b_i(v_i) = \frac{v_i}{2} \qquad \text{(p. 156) — (13.8) at } n = 2.$$

Without the linear guess (Appendix 3.2.B, pp. 157–158): with $b(\cdot)$ increasing and differentiable, $\Pr\{b_i > b(v_j)\} = b^{-1}(b_i)$, so bidder i maximizes $(v_i - b_i)\, b^{-1}(b_i)$. FOC: $-b^{-1}(b_i) + (v_i - b_i)\frac{d\, b^{-1}(b_i)}{db_i} = 0$. Impose $b_i = b(v_i)$, so $b^{-1}(b_i) = v_i$ and the derivative is $1/b'(v_i)$:

$$-v_i + \frac{v_i - b(v_i)}{b'(v_i)} = 0 \iff b'(v_i)\, v_i + b(v_i) = v_i \iff \frac{d}{dv_i}\big[b(v_i)v_i\big] = v_i.$$

Integrate: $b(v)v = \frac{1}{2}v^2 + k$; no one bids above his value and bids are nonnegative, so $b(0) = 0$, $k = 0$, $b(v) = v/2$ — the $n = 2$, $F(x) = x$ case of (13.5).

Numbers, $v = 0.8$ (illustrative): the rival bids $v_j/2$, so a bid b wins iff $v_j < 2b$.

Bid b	$\Pr\{\text{win}\} = \min(1, 2b)$	Margin $0.8 - b$	Expected payoff
0.30	0.60	0.50	0.300
0.40	**0.80**	**0.40**	**0.320**
0.50	1.00	0.30	0.300
0.80	1.00	0.00	0.000

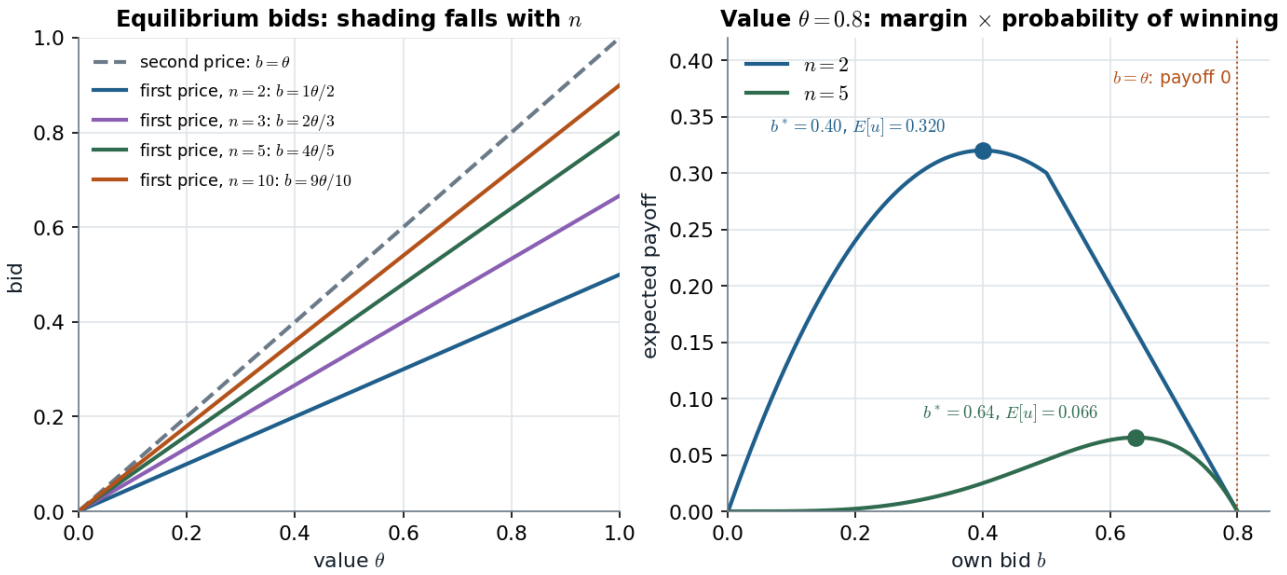


Figure — left: (13.8) for four values of n, against the second-price bid $b=\theta$. Right: (13.2) with rivals playing (13.8), $\theta=0.8$; the peaks sit at $(n-1)\theta/n$ with height θ^n/n. Sources: Tadelis pp. 277–279, Gibbons p. 156; $\theta=0.8$ illustrative.

The $n=2$ curve has a kink at $b=0.5$: above it the bidder beats every rival bid (rivals never bid more than $\frac{1}{2}$), so raising the bid only lowers the margin. With five bidders winning is rarer and the peak is lower and further right — **more competition, less shading, less surplus**.

> **Interactive companion.** The Bid Shading Desk (`companions/companion-auction-bidding.html`) — move your bid b and watch the first-price payoff tile fall on both sides of $(n-1)\theta/n$; raise n and watch the optimum slide toward θ.

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5 · C — the Dutch auction is the first-price auction (p. 279)

In a Dutch auction each bidder decides **at what price to shout "buy"**. While the price falls nothing useful is learned — the only news, "someone shouted", ends the auction. So a strategy is a map from value to a stopping price, the highest stopping price wins and pays it: **the same normal form as the first-price sealed-bid auction**. The two are **strategically equivalent** and share their BNE, (13.7).

Pair	Relation	Why
Dutch ↔ first-price	**strategically equivalent** (same normal form)	nothing observed during the descent changes anyone's choice
English ↔ second-price	**outcome-equivalent under private values** (Prop. 13.2)	in the English auction bidders *do* see others quit — harmless with private values, informative with common values

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Links

- Prev: L07-06-second-price-english · Next: the revenue comparison, L07-08-revenue-equivalence.

- Threshold and monotone strategies with continuous types: L07-04-threshold-strategies-purification.
- Lecture 1's discrete first-price auction by IESDS: `Listas/pratica-lecture-01.md` item 17 (Tadelis 4.3, p. 76) and `Listas/pratica-lecture-01-solutions/item17.tex`.
- Code: L07-12-python §7 (`fpa_bid` for any F via (13.7), `fpa_best_reply`).
- **Exercises (number + page only):** Tadelis 13.5 (p. 286), 13.6 (pp. 286–287), 13.7, 13.9 (p. 287); Gibbons 3.6 (p. 170), 3.7 (pp. 170–171); Muñoz-Garcia Ch. 8 Ex. 1 (p. 350), Ex. 2 (p. 352), Ex. 3 (p. 355), Ex. 8 (p. 371).

13.1.4 · Revenue equivalence: shading in the first-price auction exactly offsets paying the second price

Claim. With independent private values and risk-neutral bidders, **all four standard auctions give the seller the same expected revenue**. In the uniform case both the expected highest first-price bid and the expected second-highest value equal $\boxed{(n-1)/(n+1)}$ — and the equality already holds **type by type**: every bidder expects to pay the same amount in both formats. The prices are different random variables with the same mean.

Prev: L07-07-first-price-dutch · Next: L07-09-common-values-curses · Map: L07-00-map

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1 · C — the statement (p. 280)

The **revenue equivalence theorem** (Vickrey 1961; Myerson 1981; Riley and Samuelson 1981): any auction game satisfying four conditions yields the seller the **same expected revenue** and **each type of bidder the same expected payoff**:

#	Condition	Fails when …
1	each type is drawn from a "well-behaved" distribution (F_i strictly increasing and continuous)	values have atoms or gaps
2	bidders are **risk neutral**	bidders are risk averse (Tadelis exercise 13.7, p. 287)
3	the bidder with the **highest type wins**	a reserve price keeps the object unsold, or bidders are asymmetric
4	the lowest possible type $\underline{\theta}$ has expected payoff **zero**	entry fees or subsidies shift the bottom

The two easy halves (p. 279): second-price and English share a dominant-strategy outcome; first-price and Dutch are strategically equivalent. **The surprise is across the halves:** shading in the first-price auction and settling for the second-highest value in the second-price auction come to the same thing on average.

Scope. The general proof uses the envelope theorem and belongs to mechanism design; these notes **state** it and **verify** it for n uniform bidders — what Tadelis does (p. 280).

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2 · C — order statistics (pp. 280–281)

Draw $x_1, \dots, x_n$ i.i.d. from F. The **first-order statistic** $x_n^{[1]} = \max_i x_i$ is the highest draw, the **second-order statistic** $x_n^{[2]}$ the second highest.

Auction	What the seller receives	Expected revenue
second-price	the second-highest **value**	$E[\theta_n^{[2]}]$

Auction	What the seller receives	Expected revenue
first-price	the highest **bid** $s(\theta)$	$E[b_n^{[1]}]$

C.d.f. of the maximum. $\Pr\{\max_i x_i \le x\} = \Pr\{x_1 \le x\}\cdots\Pr\{x_n \le x\} = [F(x)]^n$ (independence, footnote 8, p. 281).

C.d.f. of the second highest (pp. 281–282). $\{x_n^{[2]} \le x\}$ happens in two disjoint ways: all n draws below x, or exactly one above and $n-1$ below (which one: n ways):

$$F_n^{[2]}(x) = [F(x)]^n + n\,(1-F(x))[F(x)]^{n-1} = n[F(x)]^{n-1} - (n-1)[F(x)]^n.$$

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3 · C — the first-price side (p. 281)

Each bid is $s(\theta) = \frac{n-1}{n}\theta$, so bids are uniform on $\left[0, \frac{n-1}{n}\right]$ with c.d.f. $F(b) = \frac{n}{n-1}b$. The highest bid has

$$F_n^{[1]}(x) = \Big(\frac{nx}{n-1}\Big)^n, \quad 0 \le x \le \frac{n-1}{n}, \tag{13.9}$$
$$f_n^{[1]}(x) = n\Big(\frac{n}{n-1}\Big)^n x^{n-1}.$$

$$E\big[b_n^{[1]}\big] = \int_0^{\frac{n-1}{n}} x \cdot n\Big(\frac{n}{n-1}\Big)^n x^{n-1}dx = n\Big(\frac{n}{n-1}\Big)^n \frac{1}{n+1}\Big(\frac{n-1}{n}\Big)^{n+1} = \frac{n}{n+1}\cdot\frac{n-1}{n} = \boxed{\frac{n-1}{n+1}}$$

---

4 · C — the second-price side (p. 282)

Values are uniform, $F(x) = x$:

$$F_n^{[2]}(x) = nx^{n-1} - (n-1)x^n, \tag{13.10}$$
$$f_n^{[2]}(x) = n(n-1)x^{n-2} - n(n-1)x^{n-1}.$$

$$E\big[\theta_n^{[2]}\big] = \int_0^1 (n(n-1)x^{n-1} - n(n-1)x^n)dx = \Big[(n-1)x^n - \frac{n(n-1)}{n+1}x^{n+1}\Big]_0^1 = (n-1) - \frac{n(n-1)}{n+1} = \boxed{\frac{n-1}{n+1}}$$

n	2	3	4	5	6	10
expected revenue $\frac{n-1}{n+1}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{3}{5}$	$\frac{2}{3}$	$\frac{5}{7}$	$\frac{9}{11}$

Revenue rises with n toward the top of the support, 1: competition transfers surplus to the seller.

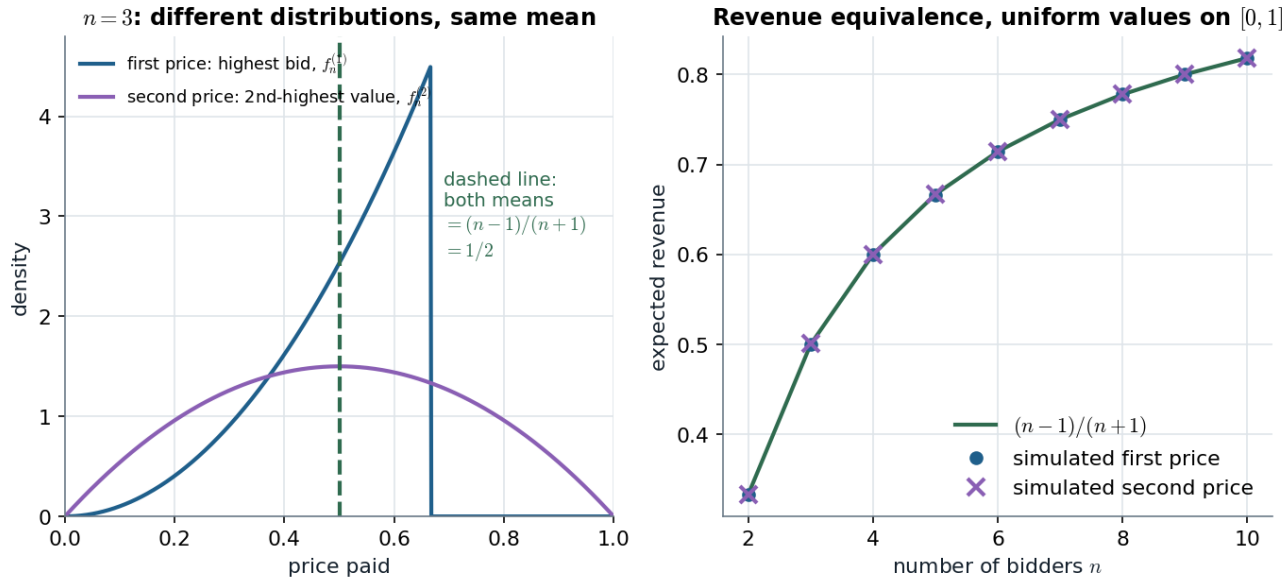


Figure — left: the two price distributions of (13.9)–(13.10) at $n = 3$ differ in shape and range and share the mean $\frac{1}{2}$. Right: $(n-1)/(n+1)$ against Monte Carlo revenues (200,000 draws per n). Source: Tadelis pp. 281–282.

The first-price price never exceeds $\frac{n-1}{n}$ and is more concentrated; the second-price price ranges over all of $[0,1]$. **Same mean, different risk:** at $n = 2$ the first-price price $\frac{1}{2}\max(\theta_1,\theta_2)$ has variance $\frac{1}{4}\cdot\frac{1}{18} = \frac{1}{72} \approx 0.014$, the second-price price $\min(\theta_1,\theta_2)$ has variance $\frac{1}{18} \approx 0.056$ (both computed in L07-12-python §8) — a reason a risk-averse *seller* might prefer the first-price format even though the means agree.

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5 · Worked — equivalence type by type

Fix a bidder of value θ facing $n-1$ uniform rivals; let Y be the highest rival value, c.d.f. y^{n-1}, density $(n-1)y^{n-2}$. Both formats allocate efficiently, so he wins iff $Y < \theta$, with probability θ^{n-1}.

	First-price	Second-price
He pays, when he wins	his bid $\frac{n-1}{n}\theta$	Y
Expected payment	$\theta^{n-1}\cdot\frac{n-1}{n}\theta$	$\int_0^\theta y\,(n-1)y^{n-2}dy = \frac{n-1}{n}\theta^n$
Expected surplus	$\theta^n - \frac{n-1}{n}\theta^n = \frac{\theta^n}{n}$	$\frac{\theta^n}{n}$

Identical for every θ — the second claim of the theorem — and the lowest type $\theta = 0$ gets 0 (condition 4). Average the payment over $\theta \sim U[0,1]$ and multiply by the n bidders:

$$n\int_0^1 \frac{n-1}{n}\theta^n\,d\theta = \frac{n-1}{n+1},$$

the revenue of §3–4 by a third route. This is L07-07-first-price-dutch §3.2 at work: the first-price bid **is** the expected second-price payment conditional on winning.

Numbers ($n = 2$, $\theta = 0.8$): wins with probability 0.8; first-price pays $0.4 \to$ expected payment 0.32; second-price expected payment $\frac{1}{2}\cdot 0.8^2 = 0.32$; surplus $0.8^2/2 = 0.32$ in both.

Interactive companion. The Bid Shading Desk (`companions/companion-auction-bidding.html`) — the "expected payment", "expected surplus" and "seller's revenue" tiles are computed once and labelled "both formats": move θ and n and they never split.

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6 · U — what the result does and does not say

- **It does not say the auctions are the same game.** Bids, the winner's payment in a given realization, and the price's variance all differ.
- **It says the format is not a revenue lever** under its four conditions (Cabral p. 141: with independent valuations, "several auction designs yield on average the same revenue level"), so the choice can rest on speed, transparency or collusion-resistance.
- **The lever is in breaking a condition**: a reserve price (condition 3), risk-averse bidders (condition 2), correlated or common values (L07-09-common-values-curses). Each is a Tadelis exercise, cited below.

Links

- Prev: L07-07-first-price-dutch · Next: L07-09-common-values-curses.
- The second-price side's dominance: L07-06-second-price-english.
- Code: L07-12-python §8 (`revenue_fpa`, `revenue_spa` by quadrature; type-by-type payments; Monte Carlo; the variance comparison).
- **Exercises (number + page only):** Tadelis 13.3 (pp. 285–286), 13.4, 13.5 (p. 286), 13.6 (pp. 286–287), 13.7 (p. 287); Muñoz-Garcia Ch. 8 Ex. 3 (p. 355), Ex. 7 (p. 371).

12.4 · 13.2 · Common values: condition on the event in which your choice matters

Claim. When the others' information bears on what an outcome is worth to you, **your action matters only in particular states — and those states are informative**. A juror is decisive only when the other juror votes to convict; a bidder wins only when the other bidder's signal was low. Ignoring that conditioning makes **voting your signal** and **bidding your expected value** fail as equilibria. Same mechanics as adverse selection: the event "trade happens" is bad news about quality.

Prev: L07-08-revenue-equivalence · Next: L07-10-formulas · Map: L07-00-map

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1 · C — the jury game (§12.4, pp. 261–264)

1.1 · Without private information

Two jurors vote Acquit (A) or Convict (C); conviction needs **both** C. The defendant is guilty (G) with prior $q > \frac{1}{2}$. Each juror gets 1 for the right verdict, 0 for the wrong one.

	A	C
A	$1-q,\ 1-q$	$1-q,\ 1-q$
C	$1-q,\ 1-q$	$q,\ q$

With $q > \frac{1}{2}$, C is weakly dominant (p. 261).

1.2 · Private signals

Each juror privately sees $\theta_i \in \{\theta_G, \theta_I\}$, independent given the truth: $\Pr\{\theta_G \mid G\} = \Pr\{\theta_I \mid I\} = p > \frac{1}{2}$. The type is **information about the state**, not a taste — and since the state enters everyone's payoff, this is a **common-values** game (p. 262). Strategies $xy \in \{AA, AC, CA, CC\}$: x after θ_G, y after θ_I.

Alone, a juror follows his signal iff $p > q$ (p. 262):

$$\Pr\{G \mid \theta_G\} = \frac{qp}{qp + (1-q)(1-p)} > q, \qquad \Pr\{G \mid \theta_I\} = \frac{q(1-p)}{q(1-p) + (1-q)p} < q,$$

and acquitting after θ_I is right iff $\Pr\{G \mid \theta_I\} < \frac{1}{2} \iff q(1-p) < (1-q)p \iff q < p$. At $p = \frac{1}{2}$ the signal is noise and the posterior stays q.

1.3 · Is "vote your signal" (CA, CA) an equilibrium? (pp. 263–264)

Signal probabilities (p. 263):

	$\theta_2 = \theta_G$	$\theta_2 = \theta_I$
$\theta_1 = \theta_G$	$qp^2 + (1-q)(1-p)^2$	$p(1-p)$
$\theta_1 = \theta_I$	$p(1-p)$	$q(1-p)^2 + (1-q)p^2$

(Off-diagonal: $qp(1-p) + (1-q)(1-p)p = p(1-p)$.)

Pivotality. Under unanimity, juror i's vote changes the verdict **only if** j **votes** C; if j plays CA, that means $\theta_j = \theta_G$. So i must evaluate his vote **conditional on** $\theta_j = \theta_G$:

$$\Pr\{G \mid \theta_i = \theta_G, \theta_j = \theta_G\} = \frac{qp^2}{qp^2 + (1-q)(1-p)^2} > q \;\Rightarrow\; C,$$

$$\Pr\{G \mid \theta_i = \theta_I, \theta_j = \theta_G\} = \frac{q(1-p)p}{p(1-p)} = q > \tfrac{1}{2} \;\Rightarrow\; C, \text{ not } A.$$

The two opposite signals **cancel**, leaving the prior. A θ_I juror who is pivotal should convict, so CA **is not a best response to** CA: voting one's signal is not a BNE.

> **The swing voter's curse** (Feddersen–Pesendorfer 1996; Austen-Smith–Banks 1996; p. 264): *"When a voter believes that his vote counts, he must condition this on the situation in which his vote counts."* It cuts against the **Condorcet jury theorem**, whose assumption that voters act as they would alone is exactly what strategic conditioning breaks.

A check before exercise 12.8 (p. 268). Tadelis states (p. 264) that both jurors playing CC is a BNE and leaves it to exercise 12.8(a). Run the pivotal logic on it: if j always convicts, i is always decisive and conditions on his own signal only — so by §1.2 a θ_I juror prefers to acquit whenever $p > q$ (e.g. $q = 0.6$, $p = 0.7$: $\Pr\{G \mid \theta_I\} = 0.391 < \frac{1}{2}$). Work the exercise keeping the case $p \le q$ versus $p > q$ explicit (L07-12-python §4 computes both).

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2 · C — the oil field and the winner's curse (§13.2, pp. 282–284)

2.1 · Pure common value

Two identical firms bid for an oil field worth the **same** v to both:

v	10	20	30
Pr	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Each firm explores and sees $\theta_i \in \{L, H\}$: $v = 10 \Rightarrow LL$; $v = 30 \Rightarrow HH$; $v = 20 \Rightarrow LH$ or HL with equal chance. Every signal pair has probability $\frac{1}{4}$ — and the signals are **correlated** through v.

2.2 · Posteriors and conditional values (pp. 283–284)

A firm with L knows the pair is LL ($v = 10$) or LH ($v = 20$), each with prior $\frac{1}{4}$: $\Pr\{\theta_j = L \mid \theta_i = L\} = \frac{1/4}{1/2} = \frac{1}{2}$. So

$$E[v \mid \theta_i = L] = \tfrac{1}{2} \cdot 10 + \tfrac{1}{2} \cdot 20 = 15, \qquad E[v \mid \theta_i = H] = \tfrac{1}{2} \cdot 20 + \tfrac{1}{2} \cdot 30 = 25.$$

2.3 · "Bid your expected value" in a second-price auction fails (p. 284)

Suppose firm 2 bids $s_2(\theta_2) = E[v \mid \theta_2]$, and firm 1 with L bids 15.

Firm 2's signal (prob.)	Firm 2's bid	Outcome for firm 1	True v	Firm 1's payoff
L $(\frac{1}{2})$	15	tie: wins with prob. $\frac{1}{2}$, pays 15	10	$\frac{1}{2}(10-15) = -2.5$
H $(\frac{1}{2})$	25	loses	20	0

$$Eu_1 = \tfrac{1}{2} \cdot \tfrac{1}{2} \cdot (10 - 15) + \tfrac{1}{2} \cdot 0 = -1.25 < 0.$$

Firm 1 would rather bid below 15 and never win. **The truthful-looking strategy is not a BNE.**

> **The winner's curse** (p. 284): *"a player wins when his signal is the most optimistic, which in the common-values setting means that he has overestimated the value of the good."* The equilibrium bid must condition on **winning**, not just on one's own signal: $E[v \mid \theta_1 = L,\ \text{I win}] = E[v \mid LL] = 10 < 15$.

The BNE of this auction is Tadelis exercise 13.8 (p. 287). The curse exists in first- and second-price formats alike (p. 284).

2.4 · Why private-value logic does not transfer

	Private values (L07-06-second-price-english)	Common values (here)
Your value	known to you: θ_i	$E[v \mid \text{all signals}]$ — you see one
Winning tells you	nothing about your value	that the rival's signal was lower
Second-price advice	bid θ_i, weakly dominant	bidding $E[v \mid \theta_i]$ loses money
English vs sealed second-price	same outcome	not the same: quitting prices reveal signals (named, Krishna 2002, Milgrom 2004)

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3 · U — the curse in the field (Cabral p. 142)

A domestic telecom operator and a foreign one bid for spectrum; values are correlated (a small market is small for both) and the domestic firm is better informed. **The uninformed bidder must shade more** — if it wins, it is probably because the informed rival bid low. Seasoned bidders learn this, so the uninformed bid less and win less often. In US offshore oil-lease auctions the winner is most often the firm already drilling the **adjacent** tract — the better-informed bidder.

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4 · U — one lesson, three settings

Setting	Your choice matters when …	… which tells you	Naive behaviour	Note
Lemons	the seller **accepts** your price	quality is below the price	pay for the average	L07-05-adverse-selection-lemons

Setting	Your choice matters when …	… which tells you	Naive behaviour	Note
Jury	the other juror **convicts**	his signal said guilty	vote your signal	§1
Common-value auction	you **win**	the rival's signal was lower	bid $E[v \mid \theta_i]$	§2

Common thread: Bayes' rule applied to the event in which your action is decisive. In each case the naive strategy is optimal for a decision taken alone and fails in the game.

Links

- Prev: L07-08-revenue-equivalence · Next: L07-10-formulas.
- Posterior beliefs from a correlated prior: L07-01-bayesian-games-harsanyi §4.
- Signaling as the informed side's attempt to escape the pool: Lecture 8.
- Rules: rules/game-theory items 5 and 19; Topic 7.
- Code: L07-12-python §4 (jury posteriors) and §9 (`cond_value`, the -1.25).
- **Exercises (number + page only):** Tadelis 12.8 (pp. 268–269), 13.8, 13.10 (p. 287); Muñoz-Garcia Ch. 7 Ex. 4 (p. 331).

Lecture 7 · Formulas — every definition, result and number, the study-guide content mapped

Prev: L07-09-common-values-curses · Next: L07-11-pitfalls · Map: L07-00-map

Estimated unit: §0 maps the study guide's block for Lecture 7, not the slides' learning goals (there are no slides yet).

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0 · The exam specification, mapped

The Lecture 7 block of `Map/study-guide.md` (title, *Before class*, *Formal side*, *Applications*), item by item:

#	Item	Note
1	Incomplete information; types; Nature; common prior; Harsanyi's transformation (Tadelis Ch. 12 opening, §12.1.1)	L07-01-bayesian-games-harsanyi §1–3
2	Beliefs from the prior by Bayes' rule (§12.1.2)	L07-01-bayesian-games-harsanyi §4
3	Strategies as type-contingent plans; Bayesian Nash equilibrium (§12.1.3)	L07-02-bayesian-nash-equilibrium §1–2
4	Solving a finite Bayesian game, ex ante and type by type (§12.2.1)	L07-02-bayesian-nash-equilibrium §3
5	A BNE with continuous actions: Cournot with private costs (Gibbons §3.1.A)	L07-03-cournot-private-costs
6	Continuous types and threshold strategies (§12.2.2)	L07-04-threshold-strategies-purification §1
7	Mixed strategies as purified BNE (§12.5; Gibbons §3.2.A)	L07-04-threshold-strategies-purification §3
8	Inefficient trade and adverse selection (§12.3, p. 258)	L07-05-adverse-selection-lemons §2–4
9	Applications: insurance, credit, Cabral's insurer (M&R p. 149; Cabral §7.4 p. 174)	L07-05-adverse-selection-lemons §5–6
10	Two-sided private information: the double auction (Gibbons §3.2.C, p. 158)	L07-05-adverse-selection-lemons §7
11	Committee voting and the swing voter's curse (§12.4)	L07-09-common-values-curses §1
12	Auction formats; IPV; second-price dominance; English auction (§13.1.1–13.1.2; Gibbons §3.2.B)	L07-06-second-price-english

#	Item	Note
13	First-price BNE and the Dutch auction (§13.1.3)	L07-07-first-price-dutch
14	Revenue equivalence (§13.1.4)	L07-08-revenue-equivalence
15	Common values and the winner's curse (§13.2; Cabral §6.4)	L07-09-common-values-curses §2–3
G	MWG §8.E (Def. 8.E.1, Prop. 8.E.1), §13.B (Def. 13.B.1, Example 13.B.1, Props. 13.B.1–2)	named in 01, 02, 04, 05

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1 · Definitions and results

Ref.	Statement	Derived / used in
Tadelis Def. 12.1 (p. 246)	Bayesian game $\langle I, \{A_i\}, \{\Theta_i\}, \{u_i(\cdot;\theta_i)\}, \{\phi_i\}\rangle$	L07-01-bayesian-games-harsanyi §3
Def. 12.2, (12.1) (p. 248)	$\Pr\{H \mid S\} = \pi(S \cap H)/\pi(S)$	01 §4
Def. 12.3 (p. 249)	pure strategy $s_i : \Theta_i \to A_i$; $\lvert A_i\rvert^{\lvert\Theta_i\rvert}$ of them	L07-02-bayesian-nash-equilibrium §1
Def. 12.4, (12.2) (p. 251)	BNE: $\forall i, \theta_i, a_i$: $\sum_{\theta_{-i}} \phi_i(\theta_{-i} \mid \theta_i) u_i(s_i(\theta_i), s_{-i}(\theta_{-i}); \theta_i) \geq \sum_{\theta_{-i}} \phi_i(\theta_{-i} \mid \theta_i) u_i(a_i, s_{-i}(\theta_{-i}); \theta_i)$	02 §2
MWG Prop. 8.E.1 (p. 256)	BNE $\Leftrightarrow$ each type with positive probability best-responds to his conditional distribution	02 §2
Gibbons (p. 145)	Cournot BNE: $q_1^* = \frac{a-2c+\theta c_H+(1-\theta)c_L}{3}$; $q_2^*(c_H) = \frac{a-2c_H+c}{3} + \frac{1-\theta}{6}(c_H - c_L)$; $q_2^*(c_L) = \frac{a-2c_L+c}{3} - \frac{\theta}{6}(c_H - c_L)$	L07-03-cournot-private-costs §3
(12.3)–(12.5) (pp. 255–257)	work iff $\theta_i \geq \sqrt{c/(1-\Pr\{s_j=1\})}$; $BR_i(\hat\theta_j) = \sqrt{c/\hat\theta_j}$ if $\hat\theta_j \geq c$, else 1	L07-04-threshold-strategies-purification §1
Claims 12.1–12.2 (pp. 256–257)	best responses are thresholds; unique BNE $\hat\theta^* = c^{1/3}$	04 §1.4
Claim 12.3 (p. 265)	perturbed Matching Pennies: H iff $\varepsilon_i \geq 0$; $\to$ mixed NE as $\varepsilon \to 0$	04 §3.2
Gibbons (3.2.1)–(3.2.2) (p. 154)	$c = p = \frac{-3+\sqrt{9+4x}}{2}$; $\Pr\{\text{Opera}\} = 1 - c/x \to \frac{2}{3}$	04 §3.3
Prop. 12.1 (p. 260)	only the lowest type can trade; any $p^* \in [10, 14]$ is a BNE price	L07-05-adverse-selection-lemons §3
Lemons ladder (this unit)	qualities $1..k$ can trade iff $\Delta \geq (k-1)s/2$	05 §4

Ref.	Statement	Derived / used in
M&R (5.1)–(5.2) (p. 151)	$P_s(x) = \frac{1}{2}(x + \bar{x})(1 + c)$, $P_b(x) = x + v$; breakdown iff $c\bar{x} > v$; mandatory coverage efficient iff $v > c\bar{x}/2$	05 §5
Gibbons (3.2.5)–(3.2.8) (pp. 160–161)	$p_b = \frac{2}{3}v_b + \frac{1}{12}$, $p_s = \frac{2}{3}v_s + \frac{1}{4}$; trade iff $v_b \geq v_s + \frac{1}{4}$	05 §7
MWG Def. 13.B.1, (13.B.6) (pp. 439, 441)	$w^* = E[\theta \mid r(\theta) \leq w^*]$	05 §8
Tadelis §12.4 (pp. 262–264)	alone: follow signal iff $p > q$; pivotal: $\Pr\{G \mid \theta_I, \theta_G\} = q$, so CA is not a BNE	L07-09-common-values-curses §1
Prop. 13.1 (p. 273)	second-price: $s_i(\theta_i) = \theta_i$ weakly dominant	L07-06-second-price-english §2
Prop. 13.2 (p. 275)	button auction: quit at $p = \theta_i$; outcome-equivalent to second-price	06 §3
Claim 13.1 (p. 276)	first-price: bidding θ_i is dominated	L07-07-first-price-dutch §1
(13.1) (p. 277)	$Eu_i = \prod_{j \neq i} F_j(s_j^{-1}(b_i))(\theta_i - b_i)$	07 §2
(13.7) (p. 278)	$s(\theta) = \theta - \int_{\underline{\theta}}^{\theta} F^{n-1}dx/F(\theta)^{n-1}$ $= E[Y \mid Y < \theta]$	07 §3
(13.8) (p. 279)	uniform: $s(\theta) = \frac{n-1}{n}\theta$; Gibbons $n = 2$: $v/2$	07 §3.1, §4
(13.9) (p. 281)	$F_n^{[1]}(x) = (nx/(n-1))^n$ on $[0, \frac{n-1}{n}]$	L07-08-revenue-equivalence §3
(13.10) (p. 282)	$F_n^{[2]}(x) = nx^{n-1} - (n-1)x^n$	08 §4
Revenue equivalence (p. 280)	four conditions ⇒ same expected revenue, same expected payoff per type; uniform: $\frac{n-1}{n+1}$	08 §1, §5
§13.2 (p. 284)	winner's curse: bidding $E[v \mid \theta_i]$ is not a BNE with common values	09 §2

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2 · Every number and worked case

Case	Inputs	Result	Note
Entry game (pp. 243–245)	$p = 2/3$ rational	row E: $(1,1), (\frac{1}{3}, \frac{4}{3}), (-\frac{1}{3}, -\frac{1}{3}), (-1, 0)$; NE $(O, FA), (O, FF), (E, AF)$; SPE (E, AF)	01 §2
Correlated prior (p. 249)	$\frac{1}{6}, \frac{1}{3}; \frac{1}{3}, \frac{1}{6}$	$\phi_1(c \mid a) = \frac{1}{3}$, $\phi_1(d \mid a) = \frac{2}{3}$	01 §4

Case	Inputs	Result	Note
Chicken (pp. 253–255)	$R = 8$, $H = 16$, $L = 0$	unique BNE (DC, dc), payoffs $(3, 3)$; swapped story: (CD, cd)	02 §3
Cournot (illustrative)	$a = 10$, $c = 2$, $c_H = 3$, $c_L = 1$, $\theta = \frac{1}{2}$	$q_1^* = \frac{8}{3}$, $q_2^* = \frac{13}{6}, \frac{19}{6}$; firm 1 profit $\frac{64}{9}$ vs $\frac{65}{9}$ with information	03 §5
Study groups (illustrative)	$c = \frac{1}{8}$	$\widehat{\theta}^* = 0.5$; $\sqrt{c} = 0.354$	04 §1.5
Purified Battle of the Sexes	$x = 1, 0.1, 0.01$	$\Pr\{\text{Opera}\} = 0.6972, 0.6703, 0.6670 \to \frac{2}{3}$	04 §3.3
Orange grove (pp. 258–260)	10/20/30 vs 14/24/34	pools 14, 19, 24; BNE prices $[10, 14]$; gains realized $\frac{4}{3}$ of 4	05 §3–4
M&R insurance (illustrative)	$\bar{x} = 100$, $c = 0.1$	market iff $v \geq 10$; mandatory efficient iff $v > 5$	05 §5
M&R bank (p. 153)	A: 1.1M sure; B: 0.9M or 1.3M	at ~10% only B borrows; B's expected profit 0.1M	05 §5
Cabral insurer (pp. 174–176)	10% high; values 20/3, costs 30/1	$p = 4.5$: -2.55 (printed -2.5); $p = 30$: 0	05 §6
Double auction (pp. 159–163)	$U[0, 1]$	gains: efficient $\frac{1}{6}$, linear $\frac{9}{64}$, one-price $x(1-x)/2 \leq \frac{1}{8}$	05 §7
MWG Example 13.B.1 (p. 442)	$\theta \sim U[0, 2]$, $r = \alpha\theta$	$\alpha > \frac{1}{2}$: $w^* = 0$	05 §8
Cabral auction (pp. 140–141)	values both 100 or both 150	posted $100 \to 100$; auction $\to 125$	06 §4
Second-price, three bidders (illustrative)	0.9, 0.6, 0.3	winner pays 0.6	06 §2.3
Gibbons first-price (illustrative $v = 0.8$)	$n = 2$	$b = 0.4$, $Eu = 0.32$; $n = 5$: $b = 0.64$, $Eu = 0.0655$	07 §4
Revenue	$n = 2, 3, 4, 5, 10$	$\frac{1}{3}, \frac{1}{2}, \frac{3}{5}, \frac{2}{3}, \frac{9}{11}$; price variance at $n = 2$: $\frac{1}{72}$ vs $\frac{1}{18}$	08
Jury (§12.4)	$q = 0.6$, $p = 0.7$	$\Pr\{G \mid \theta_I\} = 0.391$; pivotal posterior $= q$	09 §1
Oil field (pp. 283–284)	10/20/30 w.p. ¼, ½, ¼	$E[v \mid L] = 15$, $E[v \mid H] = 25$; bid 15 with L: -1.25	09 §2

Lecture 7 · Pitfalls and an original True/False self-test

Prev: L07-10-formulas · Next: L07-12-python · Map: L07-00-map

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Part I — the confusions this material produces

1 · A type is not an action, and a strategy is not an action

A strategy in a Bayesian game is a **function** $s_i : \Theta_i \to A_i$. "Firm 2 produces q_2" is not a strategy; "$q_2(c_H) = 13/6$, $q_2(c_L) = 19/6$" is. Minimal counterexample: in the entry game the incumbent has 2 actions but **4** strategies (AA, AF, FA, FF), and the equilibrium (E, AF) cannot even be written with actions alone. → L07-02-bayesian-nash-equilibrium §1.

2 · Why specify what types that were not drawn would do?

Because **the other players' expected payoffs depend on it**. Firm 1 in Gibbons's Cournot game cannot evaluate q_1 against "firm 2 produces 13/6" — it needs both quantities and θ. Same reason a strategy covers unreached nodes in Lecture 3. → L07-03-cournot-private-costs §1.

3 · Incomplete vs imperfect information

Incomplete: someone does not know another's **payoffs**. Imperfect: someone does not know the **history** of moves. Harsanyi converts the first into the second by adding Nature's move. A simultaneous-move game with known payoffs has imperfect, not incomplete, information. → L07-01-bayesian-games-harsanyi §2.

4 · Private values vs common values

Private: u_i depends on θ_i only. Common: on others' types too. Adverse selection, the swing voter's curse and the winner's curse **all need common values**; in an IPV auction learning the rival's value never changes what the object is worth to you. → L07-09-common-values-curses §2.4.

5 · "The buyer's average value exceeds the seller's, so trade happens at some average price"

False with private quality information. Orange grove: average values 24 vs 20, yet at $p = 20$ only L, M sell and that pool is worth 19 to the buyer. **The relevant average is over the sellers who accept, and it depends on the price.** → L07-05-adverse-selection-lemons §3.

6 · "Adverse selection needs a small gain from trade"

It needs a gain that is small **relative to the spread of the privately known quality**: the k-th quality joins iff $\Delta \geq (k-1)s/2$. Raising every value by the same amount (v_1) changes nothing. → L07-05-adverse-selection-lemons §4.

7 · Weakly dominant ≠ unique equilibrium, and ≠ strictly dominant

Bidding θ_i in a second-price auction is **weakly** dominant (ties in regions L and H). That selects an equilibrium; it does not prove there is no other. → L07-06-second-price-english §2.

8 · "In a first-price auction, bid your value"

Bidding your value guarantees zero (Claim 13.1). The equilibrium bid is $\frac{n-1}{n}\theta$ under uniform values — **shading falls with** n, it does not rise. → L07-07-first-price-dutch §3.

9 · Strategic equivalence vs outcome equivalence

Dutch and first-price have the **same normal form** (strategically equivalent). English and second-price give the **same outcome under private values** (Prop. 13.2) but are different games: in the English auction bidders observe exits, which matters with common values. → L07-07-first-price-dutch §5.

10 · Revenue equivalence ≠ "the auctions are the same"

Same **expected** revenue and same expected payoff per type; different bids, different realized prices, different price variance ($\frac{1}{72}$ vs $\frac{1}{18}$ at $n = 2$). And only under its four conditions. → L07-08-revenue-equivalence §4, §6.

11 · "Voting your informative signal is an equilibrium"

Under unanimity a juror is decisive only when the other votes C; conditional on that, an innocence signal leaves the posterior at $q > \frac{1}{2}$. Decision-theoretic optimality alone is not equilibrium optimality. → L07-09-common-values-curses §1.3.

12 · Adverse selection vs moral hazard

Adverse selection: **pre-contractual** hidden information (who you are). Moral hazard: **post-contractual** hidden action (what you do). M&R p. 150: a family buying maternity cover because it plans children is adverse selection; having more children *because* it is covered is moral hazard — which is outside this course. → L07-05-adverse-selection-lemons §5.

Part II — True/False self-test (original)

[**E**] easy · [**M**] medium · [**H**] hard. Each answer is settled by one paragraph.

Block A — Bayesian games and BNE

1. [**E**] In a static Bayesian game, a player's pure strategy specifies one action for each of his own types. — **T.** Def. 12.3: $s_i : \Theta_i \to A_i$. With 2 actions and 3 types there are $2^3 = 8$ pure strategies, not 2.

2. [**M**] If types are independent, a player's posterior about the others' types depends on his own type. — **F.** Independence makes $\pi(\theta_i, \theta_{-i}) = \pi_i(\theta_i)\pi_{-i}(\theta_{-i})$, so Bayes' rule returns $\phi_i(\theta_{-i} \mid \theta_i) = \pi_{-i}(\theta_{-i})$ for every θ_i. Dependence on one's own type requires correlation, as in the 1/6–1/3 prior (p. 249).

3. [**E**] Harsanyi's transformation turns a game of incomplete information into a game of imperfect information in which Nature moves first. — **T.** Nature draws types; each player observes only his own, so later movers do not know the full history (Gibbons p. 148).

4. [**M**] In the entry game with $p = 2/3$, (O, FF) is a Nash equilibrium of the induced normal form but not an equilibrium in which each type of incumbent acts optimally at his node. — **T.** Against FF the entrant prefers O ($0 > -1$); against O every incumbent strategy pays 2. But a rational incumbent reaching his node would accommodate ($1 > -1$), so FF asks him to carry out a non-credible fight.

5. [**M**] To verify a pure BNE in a finite game it suffices to check, type by type, that no single alternative action raises that type's expected payoff. — **T.** Tadelis p. 251 / MWG Prop. 8.E.1: if no type gains by changing his own action, no type-contingent rearrangement gains either.

6. [**H**] In the chicken game with $R = 8$, $H = 16$, $L = 0$, the harsh-parent type drives in the unique BNE. — **F.** The unique BNE is (DC, dc): lenient types drive, harsh types swerve. A harsh type facing dc gets $\frac{1}{2}(4-16)+\frac{1}{2}\cdot 8 = -2$ from driving, 0 from swerving.

Block B — continuous types and purification

7. [**E**] In the study-group game a student's best response to any strategy of his partner is a cutoff in his own type. — **T.** Claim 12.1: work iff $\theta_i \geq \sqrt{c/(1-\Pr\{s_j = 1\})}$, a constant cutoff.

8. [**M**] The more likely the partner is to work, the lower a student's own effort cutoff. — **F.** The cutoff $\sqrt{c/(1-\Pr\{s_j = 1\})}$ **rises** with $\Pr\{s_j = 1\}$: free-riding. BR_i is decreasing in $\hat{\theta}_j$ precisely because a lower $\hat{\theta}_j$ means j works more.

9. [**M**] With $c = 1/8$ each student works with probability one half in equilibrium. — **T.** $\hat{\theta}^* = c^{1/3} = \frac{1}{2}$ and $\Pr\{\theta_i \geq \frac{1}{2}\} = \frac{1}{2}$.

10. [**H**] Harsanyi's purification says players in a mixed equilibrium literally randomize because they are indifferent. — **F.** It says the opposite: each type plays a **pure** strict best response; the mix is the opponent's uncertainty about which type he faces (Gibbons p. 152). In Gibbons's example the probability of Opera, $1 - c/x$, tends to $2/3$ as $x \to 0$.

Block C — Cournot with private costs

11. [**M**] In Gibbons's asymmetric-cost Cournot game, the high-cost type of firm 2 produces more than it would if its cost were common knowledge. — **T.** $q_2^*(c_H)$ exceeds $(a - 2c_H + c)/3$ by $\frac{1-\theta}{6}(c_H - c_L) > 0$: firm 1, not knowing the cost is high, produces less than it would against a known high-cost rival.

12. [**H**] In that game firm 1's equilibrium quantity always equals the average of its complete- information quantities. — **F.** It does with linear demand and linear costs (p. 146), but Gibbons flags it as misleading: with total cost $c_i q_i^2$ the equality fails in general.

Block D — adverse selection and bilateral trade

13. [**E**] In Tadelis's orange-grove game, the grove trades in every BNE at a price in $[10, 14]$ when its quality is L. — **F.** Prop. 12.1 says trade *can* occur only with the lowest type, and any $p^* \in [10, 14]$ *can* be supported; it does not say trade occurs in every BNE (an offer of $p < 10$ that everybody rejects is also part of an equilibrium). Read "only if", not "if".

14. [**M**] In the orange grove, raising every seller and buyer value by 100 would let the M type trade. — **F.** The condition $\Delta \geq (k-1)s/2$ involves only the premium $\Delta = 4$ and the spread $s = 10$; the level drops out. The pool $\{L, M\}$ would be worth 119 against a price of 120.

15. [**M**] In M&R's insurance model the private market collapses whenever mandatory coverage would be inefficient. — **F.** The false converse. Collapse iff $c\bar{x} > v$; mandatory coverage is efficient iff $v > c\bar{x}/2$. In the band $c\bar{x} > v > c\bar{x}/2$ the market collapses **although** coverage is efficient.

16. [**E**] In Cabral's insurer game, a premium between the average valuation and the average cost makes money. — **F.** At $p = 4.5$ only high-risk patients buy; expected profit $0.1(4.5 - 30) = -2.55$. The averages are over the whole population; the buyers are not.

17. [M] In Gibbons's linear equilibrium of the double auction, every pair with $v_b > v_s$ trades. — **F.** Trade iff $v_b \geq v_s + \frac{1}{4}$; pairs with gains below $\frac{1}{4}$ do not trade, and expected gains are $\frac{9}{64} < \frac{1}{6}$.

Block E — auctions

18. [E] In a second-price auction with private values, bidding one's value is weakly dominant even if the bidder knows nothing about the distribution of the others' values. — **T.** Prop. 13.1's proof never uses F_j: in each case the comparison holds pointwise in the rivals' bids (p. 274).

19. [M] In a second-price auction, raising your bid above your value can change the price you pay. — **F.** Your bid only decides whether you win; the price is the highest rival bid. Overbidding changes the outcome only when that rival bid lies between your value and your bid — and then you win at a loss.

20. [E] With n bidders and values uniform on $[0, 1]$, the symmetric first-price equilibrium bid is θ/n. — **F.** It is $(n-1)\theta/n$; θ/n is the amount **shaded**. At $n = 2$: bid $\theta/2$, shade $\theta/2$ — the two coincide only there.

21. [M] The Dutch auction and the first-price sealed-bid auction have the same set of Bayesian Nash equilibria. — **T.** They are strategically equivalent: in both a strategy maps a value to a price, the highest price wins and pays it, and nothing useful is observed before one's own choice (p. 279).

22. [M] With uniform values on $[0, 1]$, the seller's expected revenue is $1/3$ with two bidders in both the first- and second-price auctions. — **T.** $(n-1)/(n+1)$ at $n = 2$; first price: $E[\frac{1}{2}\max] = \frac{1}{2} \cdot \frac{2}{3}$; second price: $E[\min] = \frac{1}{3}$.

23. [H] Revenue equivalence implies that, for each realization of values, the first- and second-price auctions raise the same revenue. — **F.** It is a statement about expectations. With values 0.9 and 0.2: first price raises 0.45, second price 0.2.

24. [H] In the type-by-type comparison, a bidder of value θ expects to pay $\frac{n-1}{n}\theta^n$ in both the first- and second-price auctions. — **T.** First price: $\theta^{n-1} \cdot \frac{n-1}{n}\theta$; second price: $\int_0^\theta y(n-1)y^{n-2}dy = \frac{n-1}{n}\theta^n$.

Block F — common values

25. [M] In the oil-field auction, a firm with a low signal that bids its expected value, 15, earns zero expected profit when the rival does the same. — **F.** It wins only on a tie with another low signal, when $v = 10$: $\frac{1}{2} \cdot \frac{1}{2}(10 - 15) = -1.25 < 0$ — the winner's curse.

26. [H] Under unanimity with informative signals, a juror who sees an innocence signal and believes the other juror votes his signal should acquit. — **F.** He is pivotal only if the other saw a guilt signal; then $\Pr\{G \mid \theta_I, \theta_G\} = q > \frac{1}{2}$ and he should convict.

Links

- Map: L07-00-map · formulas: L07-10-formulas · code: L07-12-python.
- Exam style: `rules/00_estilo_avaliacao.md` (macro-1 True/False rules).

Lecture 7 · Computation in Python

Every routine the lecture needs, each followed by the asserts that must pass. Python, numpy, scipy (quadrature only). The blocks run **in order as one script**; together they are the source of every number quoted in the notes that is not printed in a book.

Prev: L07-11-pitfalls · Map: L07-00-map · Formulas: L07-10-formulas

Reuse. Lecture 1 already checks weak dominance in the second-price auction, with a reserve price and ties, on a grid of bids (`Listas/lista-lecture-01-check.py::spa_payoff`, list Q1 b). That script runs all its checks on import, so it is cited rather than imported; §7 below only adds the three-case proof of Prop. 13.1 against a continuum of rival bids. The two companions have their own independent check, `companions/check_companions.py` (Node runs each page's model; Python recomputes it by another route).

Run. Copy the blocks into one file, or run `python Notas/L07/figures/make_figures.py` and `python Notas/L07/companions/check_companions.py` for the figure and page checks.

---

1 · Harsanyi normal form, pure equilibria, posteriors

→ L07-01-bayesian-games-harsanyi: the entry game's matrix at $p = 2/3$ and its three equilibria; the 1/6–1/3 prior.

```
import itertools
from fractions import Fraction as Fr

import numpy as np


def harsanyi_normal_form(types1, types2, prior, acts1, acts2, u1, u2):
    """Expected-payoff matrices of a two-player static Bayesian game (Tadelis Sec.
     ↪  12.1, p. 244).

    types1, types2: lists of types; prior[(t1, t2)]: common prior (dict, sums to 1);
    acts1, acts2: action lists; u1(a1, a2, t1, t2), u2(...): type-dependent payoffs.
    Returns (S1, S2, U1, U2): pure strategies (tuples, one action per type, in type
 ↪  order)
    and the ex-ante expected payoff matrices, rows = S1, columns = S2.
    """
    S1 = list(itertools.product(acts1, repeat=len(types1)))
    S2 = list(itertools.product(acts2, repeat=len(types2)))
    U1 = [[sum(prior[(t1, t2)] * u1(s1[i], s2[j], t1, t2)
               for i, t1 in enumerate(types1) for j, t2 in enumerate(types2)) for s2
                ↪  in S2] for s1 in S1]
    U2 = [[sum(prior[(t1, t2)] * u2(s1[i], s2[j], t1, t2)
               for i, t1 in enumerate(types1) for j, t2 in enumerate(types2)) for s2
                ↪  in S2] for s1 in S1]
    return S1, S2, U1, U2
```

```
def pure_nash(U1, U2):
    """All pure Nash equilibria (row, col) of a bimatrix game: mutual best
     ↪ responses."""
    R, Cn = len(U1), len(U1[0])
    return [(r, c) for r in range(R) for c in range(Cn)
            if U1[r][c] == max(U1[k][c] for k in range(R)) and U2[r][c] ==
             ↪ max(U2[r][k] for k in range(Cn))]


def posterior(prior, own_index, own_type):
    """phi_i(t_-i | t_i) by Bayes' rule (Tadelis Def. 12.2, eq. 12.1). prior[(t1,
     ↪ t2)]."""
    tot = sum(p for t, p in prior.items() if t[own_index] == own_type)
    return {t[1 - own_index]: p / tot for t, p in prior.items() if t[own_index] ==
     ↪ own_type}


# --- Entry game, Figs. 12.1-12.2 (pp. 243-245): player 2 rational (prob p) or crazy
 ↪ -------------
p = Fr(2, 3)
pay = {("O", "A"): (0, 2), ("O", "F"): (0, 2), ("E", "A"): (1, 1), ("E", "F"): (-1,
 ↪ -1)}
crazy = {("E", "F"): (-1, 2)}                       # the crazy type enjoys fighting


def entry(a1, a2, t1, t2, who):
    table = crazy if t2 == "c" and (a1, a2) in crazy else pay
    return table[(a1, a2)][who]


S1, S2, U1, U2 = harsanyi_normal_form(["one"], ["r", "c"], {("one", "r"): p, ("one",
 ↪ "c"): 1 - p},
                                      ["O", "E"], ["A", "F"],
                                      lambda a, b, s, t: entry(a, b, s, t, 0),
                                       ↪ lambda a, b, s, t: entry(a, b, s, t, 1))
assert U1[1] == [1, Fr(1, 3), Fr(-1, 3), -1] and U2[1] == [1, Fr(4, 3), Fr(-1, 3), 0]
 ↪ # matrix, p. 244
eq = {("".join(S1[r]), "".join(S2[c])) for r, c in pure_nash(U1, U2)}
assert eq == {("O", "FA"), ("O", "FF"), ("E", "AF")}
 ↪ # p. 245

# --- Posterior example, Sec. 12.1.2 (pp. 248-249)
 ↪ ------------------------------------------------
joint = {("a", "c"): Fr(1, 6), ("a", "d"): Fr(1, 3), ("b", "c"): Fr(1, 3), ("b",
 ↪ "d"): Fr(1, 6)}
assert posterior(joint, 0, "a") == {"c": Fr(1, 3), "d": Fr(2, 3)}
assert posterior(joint, 0, "b") == {"c": Fr(2, 3), "d": Fr(1, 3)}
```

2 · Chicken with incomplete information

→ L07-02-bayesian-nash-equilibrium: the full 4×4 matrix at $R = 8$, $H = 16$, $L = 0$, the unique BNE (DC, dc), and the flip to (CD, cd) when L and H swap.

```
# --- Chicken with incomplete information, Sec. 12.2.1 (pp. 252-255): R=8, H=16, L=0
 ↪  ---------------
R, K = 8, {"L": 0, "H": 16}


def chicken(i):
    def u(a1, a2, t1, t2):
        mine, theirs, k = (a1, a2.upper(), K[t1]) if i == 0 else (a2.upper(), a1,
 ↪  K[t2])
        if mine == "C":
            return 0
        return R if theirs == "C" else Fr(R, 2) - k
    return u


prior4 = {(s, t): Fr(1, 4) for s in "LH" for t in "LH"}
S1, S2, U1, U2 = harsanyi_normal_form(["L", "H"], ["L", "H"], prior4, ["C", "D"],
 ↪  ["c", "d"], chicken(0), chicken(1))
row = {"".join(s): i for i, s in enumerate(S1)}
col = {"".join(s): j for j, s in enumerate(S2)}
assert (U1[row["CD"]][col["dd"]], U2[row["CD"]][col["dd"]]) == (Fr(R, 4) - 8, Fr(3 *
 ↪  R, 4) - 4)  # R/4-H/2, 3R/4-L/4-H/4
assert [(U1[row["DC"]][c], U2[row["DC"]][c]) for c in range(4)] == [(4, 0), (3, -1),
 ↪  (3, 3), (2, 2)]
assert [(U1[row["CD"]][c], U2[row["CD"]][c]) for c in range(4)] == [(4, 0), (-1, -1),
 ↪  (-1, 3), (-6, 2)]
assert [("".join(S1[r]), "".join(S2[c])) for r, c in pure_nash(U1, U2)] == [("DC",
 ↪  "dc")]  # unique BNE, p. 254
K = {"L": 16, "H": 0}                                   # the reversed story (p. 255)
S1, S2, U1, U2 = harsanyi_normal_form(["L", "H"], ["L", "H"], prior4, ["C", "D"],
 ↪  ["c", "d"], chicken(0), chicken(1))
assert [("".join(S1[r]), "".join(S2[c])) for r, c in pure_nash(U1, U2)] == [("CD",
 ↪  "cd")]
```

3 · Cournot with a privately known cost

→ L07-03-cournot-private-costs: closed forms checked against the three FOCs and against best-response iteration; the profit comparison.

```
# --- Cournot with a privately known cost, Gibbons Sec. 3.1.A (pp. 144-146)
 ↪  ------------------------
def cournot_bne(a, c, cH, cL, theta):
    """BNE of P = a - Q; firm 1 cost c; firm 2 cost cH (prob theta) or cL. Returns
     ↪  (q1, q2H, q2L)."""
    q2H = (a - 2 * cH + c) / 3 + (1 - theta) * (cH - cL) / 6
    q2L = (a - 2 * cL + c) / 3 - theta * (cH - cL) / 6
```

```
    q1 = (a - 2 * c + theta * cH + (1 - theta) * cL) / 3
    return q1, q2H, q2L


a, c, cH, cL, th = 10.0, 2.0, 3.0, 1.0, 0.5           # illustrative: Gibbons uses
 ↪  letters only
q1, q2H, q2L = cournot_bne(a, c, cH, cL, th)
assert abs(q2H - (a - q1 - cH) / 2) < 1e-12 and abs(q2L - (a - q1 - cL) / 2) < 1e-12
 ↪  # firm 2's FOCs
assert abs(q1 - (a - c - (th * q2H + (1 - th) * q2L)) / 2) < 1e-12
 ↪  # firm 1's FOC
x = np.zeros(3)                                       # same point by best-response
 ↪  iteration (a contraction)
for _ in range(200):
    x = np.array([(a - c - (th * x[1] + (1 - th) * x[2])) / 2, (a - x[0] - cH) / 2,
 ↪  (a - x[0] - cL) / 2])
assert np.allclose(x, [q1, q2H, q2L])
assert q2H > (a - 2 * cH + c) / 3 and q2L < (a - 2 * cL + c) / 3          # Gibbons
 ↪  p. 146
assert abs(q1 - (th * (a - 2 * c + cH) / 3 + (1 - th) * (a - 2 * c + cL) / 3)) <
 ↪  1e-12      # the "misleading" average
# profits: with linear demand each margin equals the quantity, so profit =
 ↪  quantity^2
pi1_bne, pi1_ci = q1 ** 2, th * ((a - 2 * c + cH) / 3) ** 2 + (1 - th) * ((a - 2 * c
 ↪  + cL) / 3) ** 2
assert abs(pi1_bne - 64 / 9) < 1e-12 and abs(pi1_ci - 65 / 9) < 1e-12           #
 ↪  firm 1 loses from not knowing
assert q2H ** 2 > ((a - 2 * cH + c) / 3) ** 2 and q2L ** 2 < ((a - 2 * cL + c) / 3)
 ↪  ** 2
```

4 · Thresholds, purification, the jury

→ L07-04-threshold-strategies-purification and L07-09-common-values-curses §1: $c^{1/3}$ by fixed-point iteration, Gibbons's limit 2/3, the pivotal posterior $= q$, and the reply to CC.

```
# --- Study groups, Sec. 12.2.2 (pp. 255-258); MWG Example 8.E.2 (p. 256)
 ↪  --------------------------
def study_group_br(t_other, c):
    """Best-response threshold (12.5): sqrt(c / t_other) if t_other >= c, else 1."""
    return (c / t_other) ** 0.5 if t_other >= c else 1.0


for cc in (0.05, 1 / 8, 0.5, 0.9):
    t = 1.0
    for _ in range(2000):                             # iterate the symmetric map t
     ↪  -> BR(t)
        t = 0.5 * t + 0.5 * study_group_br(t, cc)     # damped: the plain map
 ↪  oscillates
    assert abs(t - cc ** (1 / 3)) < 1e-9 and cc < cc ** 0.5 < cc ** (1 / 3) < 1
```

```
# --- Purification of the Battle of the Sexes, Gibbons Sec. 3.2.A (pp. 152-154) ----------
def bos_purified(x):
    """Probability Chris plays Opera (= Pat plays Fight) in the threshold BNE: 1 - (-3 + sqrt(9+4x))/(2x)."""
    return 1 - (-3 + (9 + 4 * x) ** 0.5) / (2 * x)


for xx in (1.0, 0.1, 1e-4):
    cstar = (-3 + (9 + 4 * xx) ** 0.5) / 2            # c = p solves p^2 + 3p - x = 0 (from 3.2.1-3.2.2)
    assert abs(cstar ** 2 + 3 * cstar - xx) < 1e-12 and abs(xx / cstar - 3 - cstar) < 1e-9   # (3.2.1)
assert abs(bos_purified(1e-6) - 2 / 3) < 1e-6

# --- Jury voting, Sec. 12.4 (pp. 261-264): the pivotal posterior equals the prior ----------
for q, pp in [(0.6, 0.7), (0.55, 0.9), (0.8, 0.85)]:
    post_I = q * (1 - pp) / (q * (1 - pp) + (1 - q) * pp)
    assert (post_I < 0.5) == (pp > q)                  # alone, an innocence signal acquits iff p > q
    piv = q * (1 - pp) * pp / (q * (1 - pp) * pp + (1 - q) * pp * (1 - pp))
    assert abs(piv - q) < 1e-12                        # Pr{G | own theta_I, other theta_G} = q
# Best reply of a theta_I juror to a partner who always convicts (CC): he is always pivotal,
# so he conditions on his own signal only and acquits iff p > q (see L07-09 Sec. 1.3).
def theta_I_reply_to_CC(q, pp):
    return "A" if q * (1 - pp) / (q * (1 - pp) + (1 - q) * pp) < 0.5 else "C"


assert theta_I_reply_to_CC(0.6, 0.7) == "A" and theta_I_reply_to_CC(0.7, 0.6) == "C"
```

5 · Adverse selection

→ L07-05-adverse-selection-lemons §2–6, §8: Prop. 12.1, M&R's breakdown condition, Cabral's −2.55, MWG Example 13.B.1.

```
# --- Adverse selection, Tadelis Sec. 12.3 (pp. 258-261), Prop. 12.1 ----------
def lemons_trade(seller_vals, premium, probs=None):
    """Largest set of lowest qualities that can trade in a BNE, and the price interval supporting it.
    seller_vals sorted ascending; buyer value = seller value + premium; quality k sells iff p >= v_k.
    Returns (k_max, (p_lo, p_hi)) with k_max >= 1 counting the qualities that trade."""
    v = np.asarray(seller_vals, float)
    w = np.full(len(v), 1 / len(v)) if probs is None else np.asarray(probs, float)
```

```
    kmax = max(k for k in range(1, len(v) + 1) if np.average(v[:k] + premium,
 ↪  weights=w[:k]) >= v[k - 1] - 1e-12)
    return kmax, (v[kmax - 1], np.average(v[:kmax] + premium, weights=w[:kmax]))


k, band = lemons_trade([10, 20, 30], 4)
assert k == 1 and np.allclose(band, (10, 14))                        # only L trades,
 ↪  p* in [10, 14]
assert lemons_trade([10, 20, 30], 10)[0] == 3 and lemons_trade([10, 20, 30], 9.5)[0]
 ↪  == 2

# --- M&R's insurance market (pp. 151-152), eqs. (5.1)-(5.2): x ~ U[0, xbar]
 ↪  ----------------------
def mr_market_exists(xbar, v, c):
    """True iff some marginal buyer x has P_b(x) = x + v >= P_s(x) = (x + xbar)(1 +
     ↪  c)/2 (c < 1)."""
    xs = np.linspace(0, xbar, 10001)
    return bool(np.any(xs + v >= (xs + xbar) * (1 + c) / 2 - 1e-12))


for xbar, v, c in [(100, 5, 0.1), (100, 15, 0.1), (100, 8, 0.1)]:
    assert mr_market_exists(xbar, v, c) == (c * xbar <= v)           # breakdown iff
     ↪  c*xbar > v
# c*xbar > v > c*xbar/2: mandatory coverage would add value, the private market
 ↪  cannot exist
assert not mr_market_exists(100, 8, 0.1) and 8 > 0.1 * 100 / 2

# --- Cabral Sec. 7.4 (pp. 174-176): 10% high risk; values 20 / 3, costs 30 / 1
 ↪  ($000) -------------
def insurer_payoff(price, risk=((0.1, 20, 30), (0.9, 3, 1))):
    """Expected profit when each patient type buys iff its value >= price."""
    return sum(pr * (price - cost) for pr, val, cost in risk if val >= price)


assert abs(insurer_payoff(4.5) - (-2.55)) < 1e-12 and insurer_payoff(30) == 0     #
 ↪  Cabral rounds to -2.5
assert abs(0.1 * 20 + 0.9 * 3 - 4.7) < 1e-12 and abs(0.1 * 30 + 0.9 * 1 - 3.9) <
 ↪  1e-12

# --- MWG Example 13.B.1 (p. 442): theta ~ U[0,2], r(theta) = alpha theta
 ↪  --------------------------
def mwg_wages(alpha, grid=np.linspace(0, 2, 200001)):
    """Competitive wages w = E[theta | alpha theta <= w] = min(w / (2 alpha), 1)
     ↪  (13.B.6)."""
    return grid[np.abs(np.minimum(grid / (2 * alpha), 1.0) - grid) < 1e-9]


assert list(mwg_wages(0.75)) == [0.0]                                 # alpha > 1/2:
 ↪  complete unravelling
```

```
assert 1.0 in mwg_wages(0.4)                                    # alpha <
 ↪ 1/2: everyone hired at w = 1
```

6 · The double auction

→ L07-05-adverse-selection-lemons §7: the linear BNE, trade iff $v_b \geq v_s + \frac{1}{4}$, gains 9/64 vs 1/6 vs 1/8, and a numerical check of the buyer's best reply (3.2.5).

```
# --- Double auction, Gibbons Sec. 3.2.C (pp. 158-163)
 ↪ ------------------------------------------
def linear_double_auction(vb, vs):
    """Linear BNE (3.2.7)-(3.2.8): p_b = 2/3 v_b + 1/12, p_s = 2/3 v_s + 1/4; trade
     ↪ iff p_b >= p_s."""
    pb, ps = 2 / 3 * vb + 1 / 12, 2 / 3 * vs + 1 / 4
    return pb >= ps, (pb + ps) / 2


rng = np.random.default_rng(3)
vb, vs = rng.random(400_000), rng.random(400_000)
trade, price = linear_double_auction(vb, vs)
assert np.array_equal(trade, vb >= vs + 0.25 - 1e-12)                # trade iff
 ↪ v_b >= v_s + 1/4
assert abs(((vb - vs) * trade).mean() - 9 / 64) < 2e-3                # expected
 ↪ gains 9/64
assert abs(((vb - vs) * (vb >= vs)).mean() - 1 / 6) < 2e-3            # efficient
 ↪ benchmark 1/6
one_price = lambda x: x * (1 - x) / 2                                 # gains of
 ↪ the one-price BNE at x
assert abs(((vb - vs) * ((vs <= 0.5) & (vb >= 0.5))).mean() - one_price(0.5)) < 2e-3
 ↪ and one_price(0.5) < 9 / 64
# best-reply check of (3.2.5): buyer facing p_s = a_s + c_s v_s, here a_s = 1/4, c_s
 ↪ = 2/3
b_grid = np.linspace(0.25, 0.95, 70001)
for vbb in (0.4, 0.7, 1.0):
    obj = (vbb - 0.5 * (b_grid + (0.25 + b_grid) / 2)) * (b_grid - 0.25) / (2 / 3)
    assert abs(b_grid[np.argmax(obj)] - (2 / 3 * vbb + 1 / 3 * 0.25)) < 1e-4
```

7 · Second-price and first-price bidding

→ L07-06-second-price-english, L07-07-first-price-dutch: the three cases of Prop. 13.1; (13.7) for any F reproducing (13.8); numerical best replies.

```
# --- Second-price auction, Tadelis Prop. 13.1 (pp. 273-274)
 ↪ -------------------------------------
# Lecture 1 already checks weak dominance with a reserve price and ties on a grid of
 ↪ bids:
# Listas/lista-lecture-01-check.py::spa_payoff (Q1 b). Here: the three cases of the
 ↪ proof.
def spa_payoff(bid, theta, m):
    """Payoff against the highest rival bid m (ties: neither wins, footnote 2 p.
     ↪ 272)."""
```

```
    return np.where(bid > m, theta - m, 0.0)


m = np.linspace(0, 1, 100001)
theta = 0.6
for bid in (0.0, 0.3, 0.45, 0.75, 0.9, 1.0):
    gap = spa_payoff(theta, theta, m) - spa_payoff(bid, theta, m)
    assert gap.min() >= 0 and gap.max() > 0          # never worse, sometimes
     ↪ strictly better

# --- First-price auction, Tadelis (13.7)-(13.8) (pp. 278-279); Gibbons p. 156
 ↪ ---------------------
def fpa_bid(theta, n, F=None, lo=0.0):
    """Symmetric BNE bid (13.7): theta - int_lo^theta F(x)^(n-1) dx / F(theta)^(n-1).
     ↪ Uniform [0,1] if F is None."""
    if F is None:
        return (n - 1) * theta / n                      # (13.8)
    xs = np.linspace(lo, theta, 20001)
    return theta - np.trapezoid(F(xs) ** (n - 1), xs) / F(theta) ** (n - 1)


def fpa_best_reply(theta, n, rival_bid=lambda v, n: (n - 1) * v / n):
    """Numerical best reply to n-1 rivals using rival_bid(v, n), values uniform on
     ↪ [0,1]."""
    b = np.linspace(0, theta, 40001)
    slope = rival_bid(1.0, n)                           # linear rival strategy:
 ↪ Pr(rival bid < b) = b / slope
    U = np.minimum(1, b / slope) ** (n - 1) * (theta - b)
    return b[np.argmax(U)]


for n in (2, 3, 5, 10):
    for th in (0.2, 0.55, 0.9):
        assert abs(fpa_best_reply(th, n) - fpa_bid(th, n)) < 5e-5
        assert abs(fpa_bid(th, n, F=lambda x: x) - fpa_bid(th, n)) < 1e-8      #
         ↪ (13.7) reproduces (13.8)
assert fpa_bid(0.5, 2) == 0.25                                                  #
 ↪ Gibbons: b(v) = v/2
```

8 · Revenue equivalence

→ L07-08-revenue-equivalence: (13.9)–(13.10) by quadrature, type-by-type payments, Monte Carlo, and the variance comparison.

```
# --- Revenue equivalence, Tadelis Sec. 13.1.4 (pp. 279-282)
 ↪ ---------------------------------------
from scipy.integrate import quad


def revenue_closed(n):
    return (n - 1) / (n + 1)
```

```
def revenue_fpa(n):
    """E[highest bid]: density of the max of n bids uniform on [0,(n-1)/n], eq.
     ↪  (13.9)."""
    top = (n - 1) / n
    return quad(lambda x: x * n * (n / (n - 1)) ** n * x ** (n - 1), 0, top)[0]


def revenue_spa(n):
    """E[second-highest value]: density from (13.10), n(n-1)(x^(n-2) - x^(n-1))."""
    return quad(lambda x: x * n * (n - 1) * (x ** (n - 2) - x ** (n - 1)), 0, 1)[0]


for n in range(2, 11):
    assert abs(revenue_fpa(n) - revenue_closed(n)) < 1e-10 and abs(revenue_spa(n) -
     ↪  revenue_closed(n)) < 1e-10
# type by type: expected payment of value theta is theta^(n-1) (n-1) theta / n in
 ↪  both formats
for n in (2, 4):
    for th in (0.3, 0.8):
        pay_spa = quad(lambda x: x * (n - 1) * x ** (n - 2), 0, th)[0]          #
 ↪  E[max rival value; it < theta]
        assert abs(pay_spa - th ** (n - 1) * fpa_bid(th, n)) < 1e-12
rng = np.random.default_rng(1)
v = np.sort(rng.random((300_000, 4)), axis=1)
assert abs((3 / 4 * v[:, -1]).mean() - 0.6) < 3e-3 and abs(v[:, -2].mean() - 0.6) <
 ↪  3e-3   # n = 4: 3/5
# same mean, different risk (n = 2): Var(max/2) = 1/72, Var(min) = 1/18
var_fpa = quad(lambda x: (x / 2) ** 2 * 2 * x, 0, 1)[0] - (1 / 3) ** 2
var_spa = quad(lambda x: x ** 2 * 2 * (1 - x), 0, 1)[0] - (1 / 3) ** 2
assert abs(var_fpa - 1 / 72) < 1e-12 and abs(var_spa - 1 / 18) < 1e-12
```

9 · Common values: the oil field

→ L07-09-common-values-curses §2: $E[v \mid L] = 15$, $E[v \mid H] = 25$, and the -1.25 of bidding 15.

```
# --- Common values: the oil field, Tadelis Sec. 13.2 (pp. 283-284)
 ↪  --------------------------------
states = {(10, "L", "L"): Fr(1, 4), (20, "L", "H"): Fr(1, 4), (20, "H", "L"): Fr(1,
 ↪  4), (30, "H", "H"): Fr(1, 4)}


def cond_value(sig):
    num = sum(v * pr for (v, s1, s2), pr in states.items() if s1 == sig)
    return num / sum(pr for (v, s1, s2), pr in states.items() if s1 == sig)


assert cond_value("L") == 15 and cond_value("H") == 25                  # E[v | L] =
 ↪  15, E[v | H] = 25
```

```
# player 1 with L bids 15 while player 2 bids E[v | own signal]: tie (coin flip) only
 ↪  when both are L
bid2 = {"L": 15, "H": 25}
ev = sum(pr * (Fr(1, 2) * (v - 15) if bid2[s2] == 15 else 0)
         for (v, s1, s2), pr in states.items() if s1 == "L") / Fr(1, 2)
assert ev == Fr(-5, 4)                                                  # -1.25: the
 ↪  winner's curse
```

Links

- Map: L07-00-map · formulas: L07-10-formulas · pitfalls: L07-11-pitfalls.
- Figures: `figures/make_figures.py` (seven figures, each with its own asserts).
- Companion checks: `companions/check_companions.py`.